AstroPhysics Notes

A Pedagogical Companion

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September 7, 2026

Contents

Preface

These notes are a self-contained, pedagogically ordered exposition of the material underlying the Astrophysics course. The organization is thematic rather than following the original question order: Chapter 1 treats time systems, Chapter 2 treats coordinate systems on the celestial sphere and their transformations, Chapter 3 treats astrometric observables (parallax, proper motion, radial velocity), Chapter 4 treats stellar photometry and the magnitude system, Chapter 5 treats angular resolution in both interferometric and single-aperture telescopes, and Chapter-6 discuss Kepler problem in Astrophysics. Worked examples reproduce the numbers of the original problem sets so that the reader can check every step.

Chapter 1

Time Systems and the Rotating Earth

1.1 Two natural clocks: the solar day and the sidereal day

Every timekeeping system used in positional astronomy is built from the rotation of the Earth about its own axis, but there is more than one natural way to define "one rotation," because the direction against which the rotation is measured can be chosen in two physically distinct ways.

Definition 1.1 (Solar day) *The* solar day *is the time between two successive passages of the Sun across the local meridian (successive local solar noons). It is the day of everyday life, of length* 24^{h} *by definition (mean solar day).*

Definition 1.2 (Sidereal day) *The* sidereal day *is the time between two successive passages of a fixed direction in space – in practice, the direction to a distant star, or equivalently the vernal equinox direction* $\hat{\gamma}$ *– across the local meridian. Its length is*

$$T_{\mathrm{sid}} \simeq 23^{\mathrm{h}}56^{\mathrm{m}}4^{\mathrm{s}}.$$

Why are these two different? The Earth is not simply spinning in place: while it rotates on its axis it is also revolving around the Sun. Consider two consecutive "Days" in Figure 1.1. On Day 1 the Earth faces the Sun; the Earth then both spins on its axis *and* advances along its orbit. After one full 360° rotation relative to the distant stars (one sidereal day), the Earth has moved along its orbit by roughly

$$\frac{360^\circ}{365.25 \text{ days}} \approx 1^\circ/\text{day},$$

so the Earth must rotate through a further $\sim 1^\circ$ before the Sun is again on the local meridian. Since 360° of rotation corresponds to $24^{\mathrm{h}} = 1440^{\mathrm{m}}$, one degree of extra rotation costs

$$\frac{1440^{\mathrm{m}}}{360^\circ} = 4^{\mathrm{m}}/\text{degree},$$

so the solar day is longer than the sidereal day by about 4^{m}, i.e. $3^{\mathrm{m}}56^{\mathrm{s}}$ once the exact orbital rate is used. This is exactly the gap $24^{\mathrm{h}}00^{\mathrm{m}} - 23^{\mathrm{h}}56^{\mathrm{m}}04^{\mathrm{s}} = 3^{\mathrm{m}}56^{\mathrm{s}}$ quoted in the essential ideas of Question 1. This deficit accumulates: over one full year the Earth completes one extra sidereal rotation relative to the number of solar days, which is why a year contains 366.25 sidereal days but only 365.25 solar days.

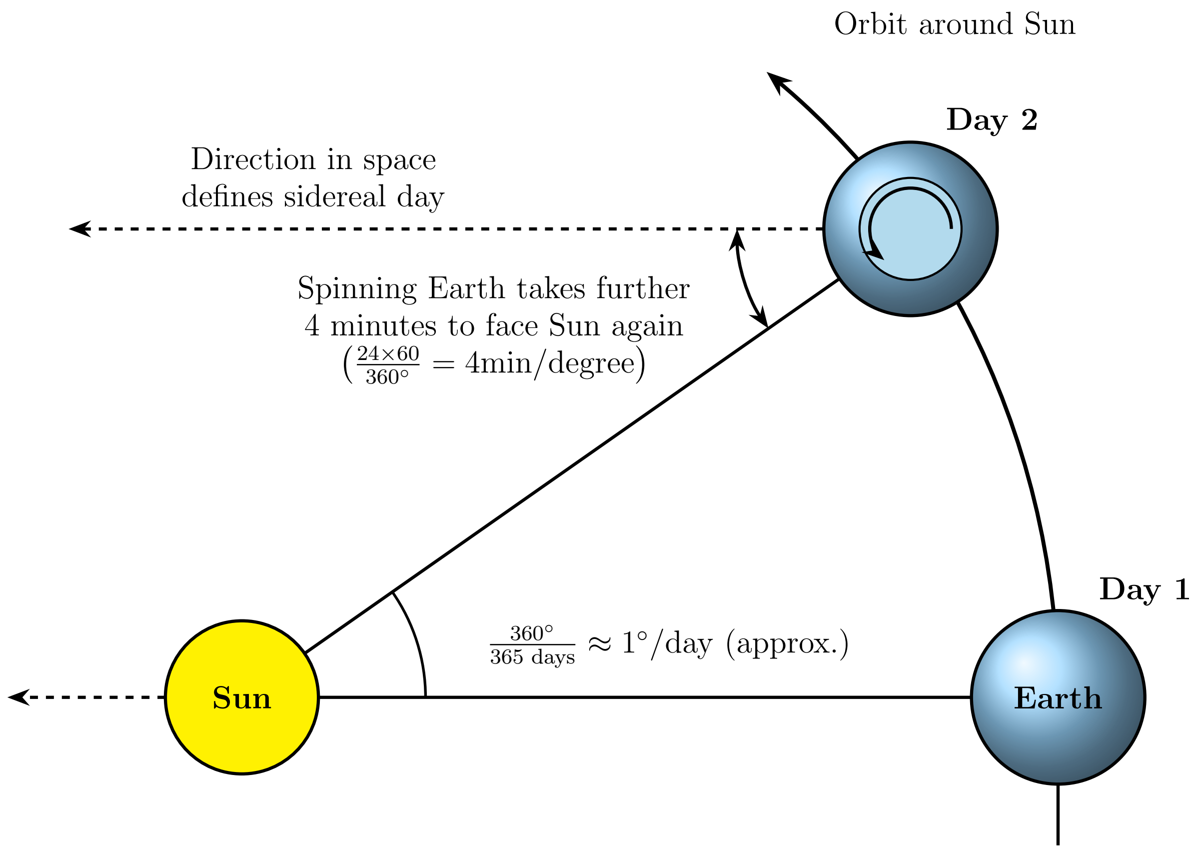


Figure 1.1: The orbital motion of the Earth around the Sun means that a full 360° rotation relative to the distant stars (a sidereal day) leaves the Earth still short of facing the Sun again; a further ~ 4 minutes of rotation are needed to complete a solar day.

Because solar time and sidereal time run at (very slightly) different rates, their numerical values coincide only at particular calendar instants (near the autumnal equinox, by convention), and afterward they steadily drift apart by about $3^{\mathrm{m}}56^{\mathrm{s}}$ per day, accumulating to a full 24^{h} offset over a year. This is precisely why a star rises about four minutes earlier each successive night.

1.2 Longitude and local time

Because the Earth rotates rigidly, all meridians (great circles through the poles) sweep past the Sun at the same angular rate

$$\omega_\oplus = \frac{360^\circ}{24^{\mathrm{h}}} = 15^\circ/\mathrm{hr} = 4^{\mathrm{m}}/\text{degree of longitude}.$$

Two observers at longitudes λ_1 and λ_2 therefore experience local solar noon at different instants of universal time, offset by

$$\Delta t = \frac{\lambda_1 - \lambda_2}{\omega_\oplus}.$$

An observer further *east* (larger longitude, using the East-positive convention) sees the Sun cross the meridian *earlier*, because the Earth rotates eastward and that meridian is swept into alignment with the Sun sooner.

Civil time versus local solar time. Civil clocks (Indian Standard Time, for example) do not track local solar time at every point in a country; instead, an entire time zone is pinned to the local solar time of a single reference meridian. India uses IST, defined at 82.5°E (Prayagraj/Allahabad meridian). Any location west of this reference meridian has its local solar noon occur *later* than 12:00 IST, since the Earth must rotate further before the Sun reaches that meridian.

Result 1.1

$$\Delta t = \frac{\lambda_{\rm ref} - \lambda_{\rm obs}}{\omega_\oplus}, \qquad \textit{Local Solar Time} = \textit{Civil Time} - \Delta t \ \textit{(if observer is west of reference)}.$$

Worked example (Problem Set 1, Q1a). IISER Thiruvananthapuram sits at $\lambda_{\rm TVM} = 77.1382°$E, west of the IST reference meridian $\lambda_{\rm IST} = 82.5°$E. The longitude offset is

$$\Delta\lambda = 82.5° - 77.1382° = 5.3618°,$$

and converting to time using $4^{\rm m}$ per degree,

$$\Delta t = 5.3618° \times 4 \text{ min}/° = 21.447 \text{ min} \approx 21^{\rm m}27^{\rm s}.$$

Because IISER TVM lags the reference meridian, its local solar noon occurs $21^{\rm m}27^{\rm s}$ *after* 12:00 IST; equivalently, when the civil clock reads 12:00:00 IST, local solar time at IISER TVM is

$$\text{LST}_\odot = 12{:}00{:}00 - 0{:}21{:}27 = 11{:}38{:}33 \text{ AM}.$$

(Here "LST" denotes local *solar* time, not to be confused with Local *Sidereal* Time introduced below, which uses the same acronym but refers to a stellar, not solar, reference.)

1.3 Local Sidereal Time (LST)

To locate objects on the celestial sphere as the Earth spins, astronomers use a sidereal clock rather than a solar one. The Local Sidereal Time at an observer's location is defined as the right ascension currently crossing the local meridian:

Result 1.2

$$LST \equiv \alpha_{\rm Meridian}.$$

Because right ascension is measured eastward from the vernal equinox direction $\hat{\gamma}$ exactly as terrestrial longitude is measured eastward from Greenwich, the LST at longitude λ relates to the sidereal time at Greenwich (Greenwich Sidereal Time, GST) exactly as local solar time relates to a reference meridian:

Result 1.3

$$LST = \alpha_{\rm Meridian} = GST + \lambda.$$

As the Earth rotates, $\alpha_{\rm Meridian}$ increases uniformly with time (by 360°, i.e. $24^{\rm h}$ of right ascension, per sidereal day), so LST behaves exactly like a clock, except that it is calibrated against the stars rather than the Sun. Since GST and GMT (solar time at Greenwich) drift apart at the sidereal-solar rate discussed above, LST and local solar time also drift apart, and one must always be careful about which "time" (solar-civil or sidereal) a given formula requires.

1.4 The right ascension of the Sun through the year

The Sun is not a fixed point on the celestial sphere: because the Earth orbits it, the Sun's apparent position among the distant stars (its *ecliptic* path) changes throughout the year, so its equatorial coordinates $(\alpha_\odot, \delta_\odot)$ are functions of date, unlike a star's (α, δ), which is essentially fixed (its motion is entirely due to the Earth's rotation, not any true motion of the star, over the timescale of an observation). The Sun's celestial coordinates at the four cardinal points of the year are

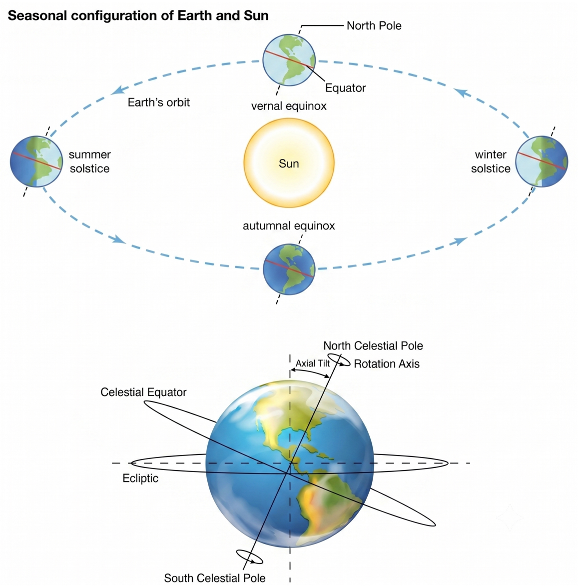


Figure 1.2: The Sun is on the Tropic of Cancer at the summer solstice and on the Tropic of Capricorn at the winter solstice, and on the equator at the equinoxes.

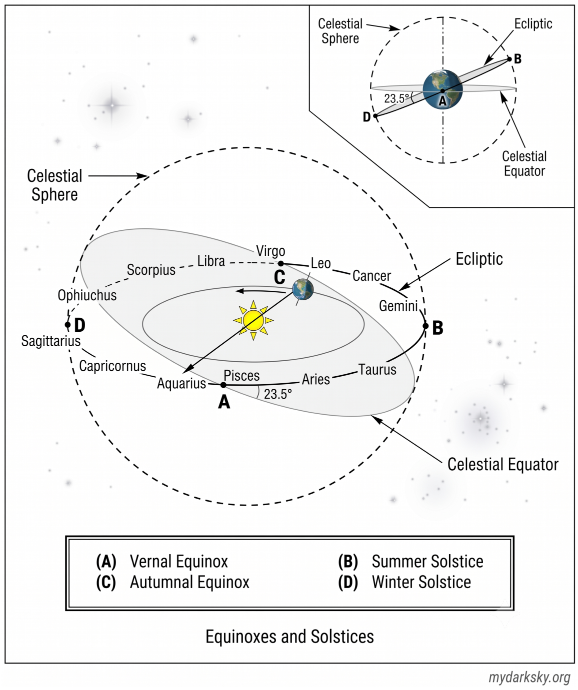


Figure 1.3: The Sun's right ascension and declination change throughout the year as it traces the ecliptic; at the March equinox the Sun lies in the direction of the vernal equinox.

$$
\begin{aligned}
\text{March (vernal) equinox:} &\quad \alpha_\odot = 0^{\rm h}, &\delta_\odot = 0^\circ, &\quad \alpha_{\rm GM} = 0^{\rm h},\\
\text{June (summer) solstice:} &\quad \alpha_\odot = 6^{\rm h}, &\delta_\odot \simeq +23.4^\circ, &\quad \alpha_{\rm GM} = 6^{\rm h},\\
\text{September (autumnal) equinox:} &\quad \alpha_\odot = 12^{\rm h}, &\delta_\odot = 0^\circ, &\quad \alpha_{\rm GM} = 12^{\rm h},\\
\text{December (winter) solstice:} &\quad \alpha_\odot = 18^{\rm h}, &\delta_\odot \simeq -23.4^\circ, &\quad \alpha_{\rm GM} = 18^{\rm h}.
\end{aligned}
$$

The right ascension of the Greenwich meridian, $\alpha_{\rm GM}$, is by definition equal to GST. The boxed identity used repeatedly in Problem Set 1,

$$\alpha_{\rm GM} = \alpha_\odot \quad \text{when the Sun crosses the Greenwich meridian,}$$

is simply the statement that GST equals the Sun's right ascension precisely at the moment of solar noon at Greenwich – the sidereal clock and the Sun's own coordinate agree exactly once per day, at local solar noon, by the very definition of local sidereal time being the right ascension on the meridian.

1.5 Mean and apparent sidereal time

The treatment above quietly assumed that the vernal equinox $\hat{\gamma}$ is a perfectly fixed direction in space, which is only approximately true. The Earth's rotation axis does not maintain a constant orientation: it sweeps out a slow cone (*precession*, Section 1.7 below) and wobbles about that cone with smaller, faster oscillations (*nutation*) driven by the changing torques of the Moon and Sun on the Earth's equatorial bulge as they move along their own orbits. Consequently, standard positional-astronomy practice, as fixed by International Astronomical Union convention and implemented in the U.S. Naval Observatory's almanac routines and in Meeus's widely used *Astronomical Algorithms*, distinguishes two different equinox directions:

Definition 1.3 (Mean and true equinox) *The* mean equinox *is the equinox direction corrected for precession only (a smoothly, predictably drifting direction). The* true equinox *is additionally corrected for nutation (the shorter-period wobble on top of precession).*

Sidereal time referenced to the mean equinox is *mean sidereal time* (GMST at Greenwich, LMST locally); sidereal time referenced to the true equinox is *apparent sidereal time* (GAST, LAST). The two differ by the *equation of the equinoxes*,

$$\mathrm{GAST} - \mathrm{GMST} = \Delta\psi \cos\epsilon \equiv \mathrm{eqeq},$$

where $\Delta\psi$ is the nutation in longitude (a sum of many periodic terms, dominated by an 18.6-year term tied to the precession of the Moon's orbital nodes) and ϵ is the obliquity of Section 2.6. The equation of the equinoxes never exceeds about 1.2 s in size, which is why Problem Set 1 (and indeed most introductory treatments) can safely use *mean* sidereal time throughout without comment: the correction is negligible at the precision – arcseconds to arcminutes – of a hand calculation, but it is not negligible for professional-grade telescope pointing or spacecraft navigation, which is why the full apparatus of mean versus apparent quantities exists at all.

1.6 Precession of the equinoxes and reference epochs

Because the direction $\hat{\gamma}$ that defines the zero point of both right ascension and sidereal time is itself slowly moving, a star's quoted (α, δ) silently carries an implicit *epoch*: the date for which that particular equinox direction was used to set up the coordinate grid. This is why star catalogs and planetarium software always specify an epoch, most commonly the standard epoch **J2000.0** (12:00 Terrestrial Time on 1 January 2000).

Definition 1.4 (Precession) Lunisolar precession *is the slow, steady conical sweep of the Earth's rotation axis (and hence of the celestial pole and equator) caused by the gravitational torque that the Moon and Sun exert on the Earth's equatorial bulge, which is not perfectly aligned with the orbital plane. The intersection of the (moving) celestial equator with the (essentially fixed) ecliptic – i.e. the equinox direction itself – consequently drifts westward along the ecliptic.*

The present-day rate of this drift is approximately

$$\dot{\lambda}_{\mathrm{eq}} \simeq 50.3''/\mathrm{yr},$$

corresponding to a full 360° circuit (a return of the equinox to the same point among the stars) in about 25,772 years – the famous "Platonic year," first inferred by Hipparchus around 129 BCE from a comparison of his own star positions with older Babylonian records. Over the several decades typically spanned by a research career, precession alone shifts a star's right ascension and declination by tens of arcseconds, comparable to or larger than many of the angular quantities computed in Chapter 2; this is why any coordinate taken from an older catalog must be precessed to the epoch of the observation before it is used in a formula such as Eq. (2.7), and why this document, like the problem sets it accompanies, implicitly treats all quoted (α, δ) as already referring to a common, stated epoch.

Chapter 2

Coordinate Systems on the Celestial Sphere

2.1 The equatorial coordinate system

The celestial sphere is an imaginary sphere of arbitrary radius, centered on the observer (or on the Earth), on which all directions in the sky can be represented as points. The *equatorial coordinate system* extends the Earth's own latitude-longitude grid onto the sky: the celestial equator is the projection of the Earth's equator, the celestial poles are the projections of the Earth's rotation axis, and a point on the sphere is labeled by

Definition 2.1 (Right ascension and declination) Declination δ *is the angle north (+) or south (−) of the celestial equator, playing the role of latitude.* Right ascension α *is the angle measured eastward along the celestial equator from the vernal equinox direction* $\hat{\gamma}$*, playing the role of longitude. By convention* α *is expressed in hours, minutes, and seconds of time* ($24^{\mathrm{h}} = 360°$)*, while* δ *is expressed in degrees.*

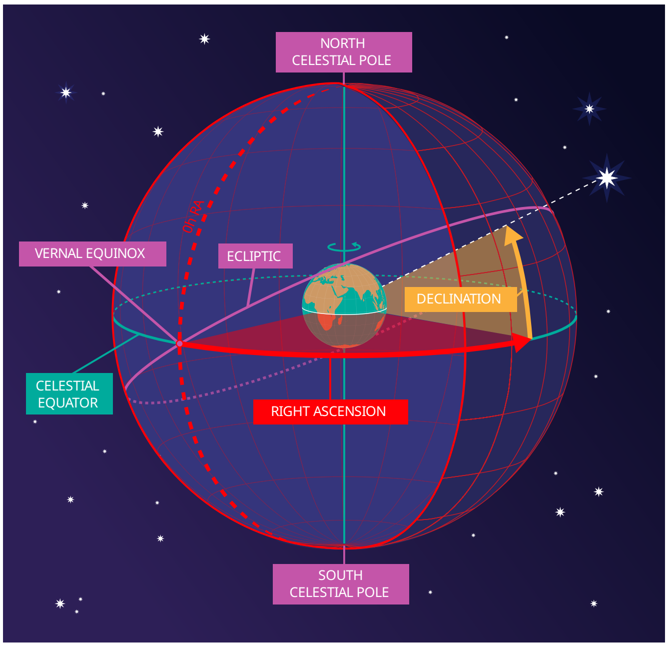


Figure 2.1: The z-axis of the equatorial Cartesian frame is the Earth's rotation axis; the equatorial plane is the xy-plane. A point P on the celestial sphere is labeled by (α, δ).

Placing the origin at the observer (or the Earth's center) with the z-axis along the rotation axis and the x-axis toward the vernal equinox, a source at (α, δ) has the Cartesian unit vector

Result 2.1

$$\hat{\mathbf{r}}_{\text{eq}} = \begin{pmatrix} \cos\delta\cos\alpha \\ \cos\delta\sin\alpha \\ \sin\delta \end{pmatrix}.$$

This is nothing but the ordinary spherical-to-Cartesian dictionary with δ playing the role of latitude (measured from the equatorial plane, hence $\sin\delta$ for the z-component rather than $\cos\delta$) and α playing the role of the azimuthal angle in that plane.

Because the equatorial system is tied to the celestial sphere – to the distant stars, via the fixed direction $\hat{\gamma}$ and the fixed pole – and not to the spinning Earth, a star's (α, δ) is very nearly constant. The apparent nightly motion of stars across the sky is entirely a reflection of the Earth's rotation carrying the observer's local meridian and horizon around underneath a fixed pattern of stars.

2.2 The horizontal coordinate system and the hour angle

The *horizontal* (alt-az) system is instead tied to the observer's own local vertical and horizon, and therefore co-rotates with the Earth (and hence with the observer).

Definition 2.2 (Altitude, azimuth, hour angle) Altitude *h is the angle above the local horizon (with $h = 90°$ at the zenith);* azimuth *A is the angle around the horizon, usually measured from north through east. The* hour angle *H of an object is the angle, measured along the celestial equator, from the local meridian to the object's hour circle, defined by*

$$H \equiv \alpha_{\text{Meridian}} - \alpha = LST - \alpha,$$

with $H > 0$ once the object has crossed the meridian and is moving toward the western horizon.

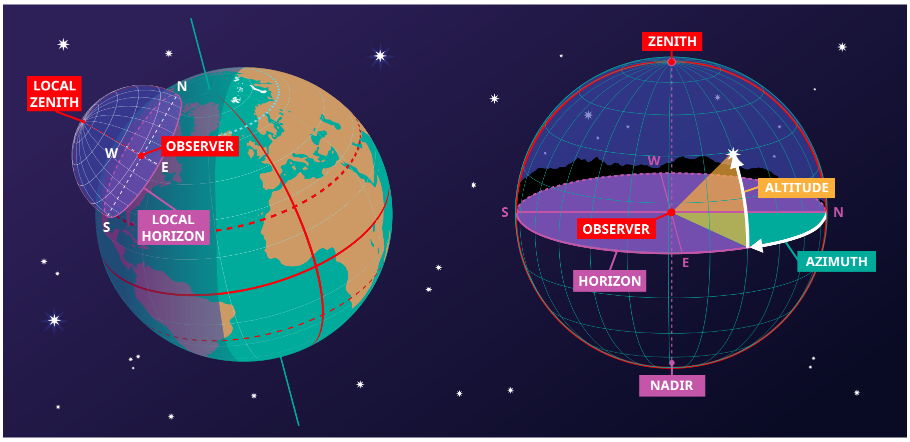


Figure 2.2: The horizontal coordinate system: the great circle through the north point, zenith, and south point is the local meridian. Projected onto the celestial sphere, the local meridian and a fixed longitude line coincide.

The hour angle is, in effect, the "local, corotating" version of right ascension: while α is measured from the fixed direction $\hat{\gamma}$, H is measured from the (rotating) local meridian. As the Earth turns, H increases uniformly for a fixed star, exactly tracking the passage of time – this is precisely why LST (the meridian's right ascension) is a valid clock.

2.3 Deriving the horizontal ↔ equatorial transformation

We now derive, from first principles, the relationship between (h, A) and (δ, H) used throughout Problem Set 1, Question 2. The key idea is to write the unit vector toward a source in *two different orthonormal bases* that share the same origin, and to relate the two bases by a rotation.

2.3.1 The source vector in a meridian-aligned equatorial frame

Instead of aligning the x-axis with the vernal equinox (as in Section 2.1), align it instead with the *local meridian* at the instant of interest – i.e., let the 0^{h} hour circle be the current meridian. In this frame the source's angular coordinate along the equator is $\alpha - \alpha_{\mathrm{Meridian}} = -H$, so

$$\hat{\mathbf{s}}_{\mathrm{eq}} = \begin{pmatrix} \cos\delta\cos(\alpha - \alpha_{\mathrm{Meridian}}) \\ \cos\delta\sin(\alpha - \alpha_{\mathrm{Meridian}}) \\ \sin\delta \end{pmatrix} = \begin{pmatrix} \cos\delta\cos H \\ -\cos\delta\sin H \\ \sin\delta \end{pmatrix}. \tag{2.1}$$

This is simply Eq. (2.1) of Section 2.1 with α replaced by $-H$, using $\cos(-H) = \cos H$ and $\sin(-H) = -\sin H$.

2.3.2 The local horizontal basis, expressed in the same frame

We now ask: what are the equatorial-frame coordinates of the three cardinal directions of the horizontal system – east, zenith, and north – for an observer at latitude ϕ_0?

- **East** points along the celestial equator, 90° west of the meridian: $(\delta, H) = (0, -90°)$.
- **Zenith** is directly overhead, on the meridian, at the observer's own declination-equivalent: $(\delta, H) = (\phi_0, 0)$.
- **North** (the horizontal-system north point, at the intersection of the meridian and horizon) is 90° from the zenith, along the meridian, giving $(\delta, H) = (90° + \phi_0, 0)$.

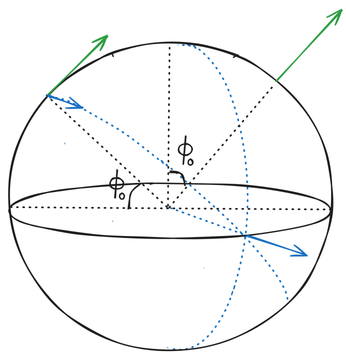


Figure 2.3: Local North and East directions.

Substituting each of these into Eq. (2.1) (using $\cos(90° + \phi_0) = -\sin\phi_0$ and $\sin(90° + \phi_0) = \cos\phi_0$ for the north vector) gives

$$\hat{\mathbf{n}} = \begin{pmatrix} -\sin\phi_0 \\ 0 \\ \cos\phi_0 \end{pmatrix}, \qquad \hat{\mathbf{e}} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \qquad \hat{\mathbf{z}} = \begin{pmatrix} \cos\phi_0 \\ 0 \\ \sin\phi_0 \end{pmatrix}. \tag{2.2}$$

One should check that $\{\hat{\mathbf{n}}, \hat{\mathbf{e}}, \hat{\mathbf{z}}\}$ is indeed an orthonormal, right-handed set: $\hat{\mathbf{n}} \cdot \hat{\mathbf{e}} = 0$, $\hat{\mathbf{e}} \cdot \hat{\mathbf{z}} = 0$, $\hat{\mathbf{n}} \cdot \hat{\mathbf{z}} = -\sin\phi_0\cos\phi_0 + \cos\phi_0\sin\phi_0 = 0$, and each has unit norm. Geometrically, Eq. (2.2) is just a rotation of the equatorial basis $\{\hat{x}, \hat{y}, \hat{z}\}$ about the (common) y-axis by the co-latitude $90° - \phi_0$; this is the rotation that tilts the celestial pole down to the observer's local zenith-horizon geometry as one moves from the pole to latitude ϕ_0.

2.3.3 Assembling the transformation

Any unit vector can be decomposed along a horizontal basis using the usual spherical dictionary, with altitude h playing the role of "latitude" above the horizon plane and azimuth A the role of "longitude" within that plane, measured from north through east:

$$\hat{\mathbf{s}}_{\rm eq} = (\cos h \cos A)\,\hat{\mathbf{n}} + (\cos h \sin A)\,\hat{\mathbf{e}} + (\sin h)\,\hat{\mathbf{z}}.$$

Writing this as a matrix acting on the column vector of horizontal components, using the basis vectors of Eq. (2.2) as its columns,

$$\hat{\mathbf{s}}_{\rm eq} = \begin{pmatrix} -\sin\phi_0 & 0 & \cos\phi_0 \\ 0 & 1 & 0 \\ \cos\phi_0 & 0 & \sin\phi_0 \end{pmatrix} \begin{pmatrix} \cos h \cos A \\ \cos h \sin A \\ \sin h \end{pmatrix}. \tag{2.3}$$

Because the matrix above is a rotation matrix, its inverse equals its transpose, so Eq. (2.3) inverts to

$$\begin{pmatrix} \cos h \cos A \\ \cos h \sin A \\ \sin h \end{pmatrix} = \begin{pmatrix} -\sin\phi_0 & 0 & \cos\phi_0 \\ 0 & 1 & 0 \\ \cos\phi_0 & 0 & \sin\phi_0 \end{pmatrix} \begin{pmatrix} \cos\delta \cos H \\ -\cos\delta \sin H \\ \sin\delta \end{pmatrix}, \tag{2.4}$$

using Eq. (2.1) for $\hat{\mathbf{s}}_{\rm eq}$ on the right. Multiplying out Eq. (2.4) row by row gives the three scalar relations

$$\cos h \cos A = -\sin\phi_0 \cos\delta \cos H + \cos\phi_0 \sin\delta, \tag{2.5}$$

$$\cos h \sin A = -\cos\delta \sin H, \tag{2.6}$$

$$\sin h = \cos\phi_0 \cos\delta \cos H + \sin\phi_0 \sin\delta. \tag{2.7}$$

Equation (2.7) is the master formula for altitude, and dividing the second relation by the first gives azimuth via a quadrant-sensitive tangent,

$$\tan A = \frac{-\cos\delta \sin H}{-\sin\phi_0 \cos\delta \cos H + \cos\phi_0 \sin\delta} = \frac{\sin H}{\sin\phi_0 \cos H - \cos\phi_0 \tan\delta}. \tag{2.8}$$

Worked example (Sirius at LST $22^{\rm h}00^{\rm m}$, IISER TVM). With $\delta = -16.716°$, $\phi_0 = 8.6833°$, $\alpha = 06^{\rm h}45^{\rm m}08.917^{\rm s}$, and LST $= 22^{\rm h}00^{\rm m}00^{\rm s}$,

$$H = \mathrm{LST} - \alpha = 15^{\rm h}14^{\rm m}51^{\rm s} \approx 228.71°.$$

From Eq. (2.7),

$$\sin h = \sin(8.6833°)\sin(-16.716°) + \cos(8.6833°)\cos(-16.716°)\cos(228.71°) = -0.668,$$

so $h \approx -41.9°$ (Sirius is well below the horizon at this LST). From Eq. (2.8),

$$\tan A = \frac{\sin(228.71°)}{\sin(8.6833°)\cos(228.71°) - \cos(8.6833°)\tan(-16.716°)},$$

which gives a reference angle of 75.26°; since $\cos h \cos A < 0$ and $\cos h \sin A < 0$ place the true azimuth in the third quadrant, $A = 180° + (-75.26°) = 104.74°$ measured appropriately – concretely $A \approx 104.74°$, i.e. slightly east of due south-west, consistent with a source that has set below the horizon on the western side.

2.4 Rising and setting

An object rises or sets when its altitude passes through $h = 0°$. Setting $h = 0$ in Eq. (2.7),

$$0 = \cos\phi_0 \cos\delta \cos H + \sin\phi_0 \sin\delta \quad \Longrightarrow \quad \boxed{\cos H = -\tan\phi_0 \tan\delta.} \tag{2.9}$$

This equation has two solutions, $H = \pm H_0$ with $H_0 = \arccos(-\tan\phi_0 \tan\delta)$, corresponding respectively to setting ($H = +H_0$, object moving west past the horizon) and rising ($H = -H_0$, object about to cross from below the horizon into view in the east). The corresponding sidereal times follow immediately from $\mathrm{LST} = \alpha + H$.

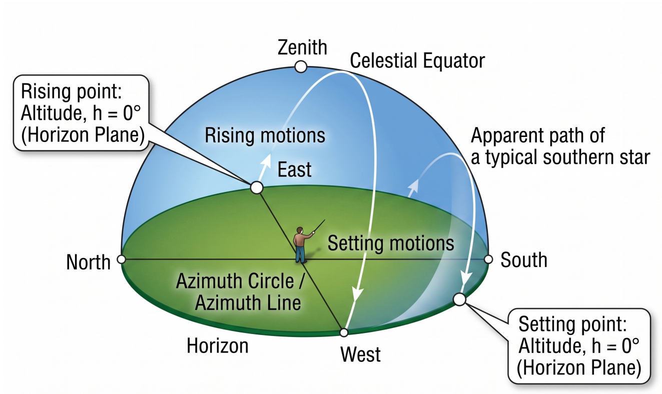


Figure 2.4: As the celestial sphere turns, an object traces a small circle of constant declination. It rises in the east and sets in the west when this circle crosses the horizon, $h = 0°$.

Worked example (Sirius, IISER TVM). With $\phi_0 = 8.6833°$ and $\delta = -16.716°$,

$$\cos H = -\tan(8.6833°)\tan(-16.716°) = 0.0458 \implies H = \pm 87.37° = \pm 5^{\mathrm{h}}49^{\mathrm{m}}29^{\mathrm{s}}.$$

With $\alpha_{\mathrm{Sirius}} \approx 06^{\mathrm{h}}45^{\mathrm{m}}09^{\mathrm{s}}$,

$$\text{Rise LST} = \alpha - H_0 = 00^{\mathrm{h}}55^{\mathrm{m}}40^{\mathrm{s}}, \qquad \text{Set LST} = \alpha + H_0 = 12^{\mathrm{h}}34^{\mathrm{m}}38^{\mathrm{s}}.$$

Note that $\cos H = -\tan\phi_0 \tan\delta$ has no real solution when $|\tan\phi_0 \tan\delta| > 1$: this is the familiar statement that, at sufficiently high latitude, a sufficiently far-north or far-south star never rises or never sets (circumpolar behavior), a case that does not arise near the Earth's equator, where $\tan\phi_0$ is small.

2.5 The zenith angle at local solar noon

Definition 2.3 (Zenith angle) *The* zenith angle z *of an object is its angular distance from the observer's zenith,* $z = 90° - h$.

At local solar *noon* the Sun is, by definition, on the observer's meridian, so $H_\odot = 0$. Substituting $H = 0$ into the master altitude formula, Eq. (2.7),

$$\sin h_\odot = \cos\phi_0 \cos\delta_\odot + \sin\phi_0 \sin\delta_\odot = \cos(\phi_0 - \delta_\odot),$$

using the cosine-difference identity. Since $\sin h = \cos z$, this gives the strikingly simple noon relation

Result 2.2

$$z_{\mathrm{noon}} = |\phi_0 - \delta_\odot|.$$

This result also has a direct geometric derivation, shown in Figure 2.5: at noon the Sun's rays are parallel lines striking both the subsolar point (latitude $\phi_s = \delta_\odot$, the point where the Sun is directly overhead) and the observer (latitude ϕ_0). Because both points lie on the same meridian at local noon, the angle between the observer's zenith direction and the direction to the Sun equals the difference in their latitudes on the Earth, precisely $|\phi_0 - \delta_\odot|$, by simple properties of parallel lines cut by radii of a common circle (this is the same construction used to explain Eratosthenes' measurement of the Earth's circumference).

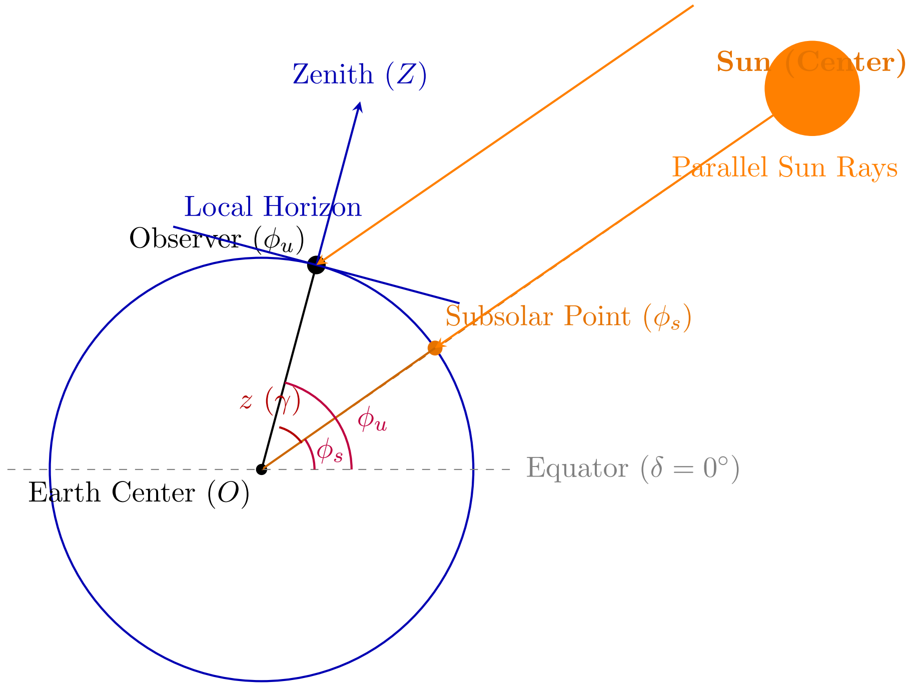


Figure 2.5: At local solar noon, parallel rays from the Sun strike the subsolar point and the observer. Since both are on the same meridian, the zenith angle at the observer equals the absolute difference of the two latitudes, $z = |\phi_u - \phi_s|$.

At the summer solstice, $\delta_\odot = +23.44°$ (the Tropic of Cancer). For IISER TVM, $\phi_0 = 8.6833°$N,

$$z_{\text{noon}} = |8.6833° - 23.44°| = 14.76°.$$

Physically, since IISER TVM's latitude is south of the Tropic of Cancer, the Sun at the summer solstice noon passes slightly *north* of the zenith there, giving a modest zenith angle rather than the Sun being directly overhead (which happens only for observers exactly at $\phi_0 = 23.44°$ on that date, or more generally whenever $\phi_0 = \delta_\odot$).

2.6 From true position to observed position: refraction and aberration

Everything derived so far relates an object's cataloged, geometric coordinates to a *true* altitude and azimuth. Two further physical effects intervene between the true position and what a telescope or the eye actually measures, and classical spherical-astronomy textbooks (Smart's *Textbook on Spherical Astronomy*; W. M. Green's *Spherical Astronomy*) treat both alongside precession and nutation as standard reductions applied to every observation.

Definition 2.4 (Atmospheric refraction) *Because the Earth's atmosphere has a refractive index that decreases smoothly with height, a light ray from a star is bent toward the local vertical as it descends through the atmosphere, so every object appears* higher *above the horizon than its true (geometric) altitude.*

For altitudes not too close to the horizon, refraction is well approximated by

$$R \simeq 58.2'' \cot h,$$

which already shows the qualitative behavior: refraction vanishes at the zenith ($h = 90°$) and grows without bound as $h \to 0$. Near the horizon this simple cotangent law breaks down (a ray grazing the horizon samples a long, curved path through the densest part of the atmosphere), and one instead uses empirical fitting formulas such as Bennett's (1982) formula,

$$R[\text{arcmin}] \simeq \frac{1.02}{\tan\left(h + \dfrac{10.3°}{h + 5.11°}\right)},$$

which is calibrated to remain accurate all the way down to the horizon and reduces to a refraction of about $34'$ (roughly the Sun's own angular diameter) for an object exactly on the horizon – the everyday consequence being that the Sun is often still geometrically *below* the horizon at the moment it appears to touch it at sunset. Refraction depends on local air pressure and temperature and is corrected for in precise work by including the factor $(P/1010\,\text{hPa}) \times (283/(273{+}T[°\text{C}]))$. Because refraction was neglected throughout Sections 2.3–2.4, the altitudes and rise/set times computed there (e.g. for Sirius) are, strictly, *true* altitudes and times; the visually observed rise would occur a few minutes earlier, and the observed set a few minutes later, than the geometric values found from Eq. (2.9).

Definition 2.5 (Aberration of starlight) Stellar aberration *is an apparent shift in a star's position caused by the combination of the finite speed of light with the observer's own motion, exactly analogous to the way rain appears to fall at a slanted angle to someone running through it even when it is actually falling vertically.*

For an observer moving at speed v transverse to the line of sight, the apparent shift in position has magnitude of order v/c radians. For the Earth's annual orbital motion around the Sun, $v \simeq 29.8$ km/s, giving the *constant of aberration*

$$\kappa = \frac{v}{c} \simeq 20.5''.$$

This is roughly a hundred times larger than a typical trigonometric parallax, which is precisely why aberration (discovered by James Bradley in 1727 as an unwanted signal while he was searching for stellar parallax) was detected first, and only later disentangled from the true parallactic motion, which has a different phase and a star-dependent amplitude, whereas κ is the same $20.5''$ for every star on the sky. A smaller diurnal aberration, of order $0.3''$ and set by the observer's own rotational speed around the Earth's axis rather than the Earth's orbital speed, is a further refinement used in the most precise astrometric work.

Both corrections matter for exactly the same conceptual reason: the equatorial coordinates (α, δ) tabulated in a star catalog are *mean, extra-atmospheric* coordinates, while any real measurement of (h, A) or (δ, H) is unavoidably an *apparent, atmosphere-affected* quantity; reconciling the two is the subject matter of the "reduction of observations" chapters in every classical positional-astronomy textbook, and refraction and aberration are, respectively, the terrestrial and kinematic pieces of that reduction.

2.7 Angular distance between two points on a sphere

A recurring calculation in positional astronomy is the angular separation between two points specified by a pair of spherical angles, whether they are two stars given in (α, δ) or two ground stations given in (ϕ, λ). The technique is identical in both cases and rests on a single geometric fact.

Result 2.3 *If $\hat{\mathbf{r}}_1$ and $\hat{\mathbf{r}}_2$ are unit vectors toward two points on a sphere, the angle θ between the directions (equivalently, the great-circle angular separation of the two* *points as seen from the sphere's center) satisfies*

$$\cos\theta = \hat{\mathbf{r}}_1 \cdot \hat{\mathbf{r}}_2.$$

This is simply the definition of the dot product, $\hat{\mathbf{r}}_1 \cdot \hat{\mathbf{r}}_2 = |\hat{\mathbf{r}}_1||\hat{\mathbf{r}}_2|\cos\theta$, applied to unit vectors. Writing out both vectors using the equatorial representation of Section 2.1,

$$\hat{\mathbf{r}}_i = \begin{pmatrix} \cos\delta_i \cos\alpha_i \\ \cos\delta_i \sin\alpha_i \\ \sin\delta_i \end{pmatrix}, \qquad i = 1, 2,$$

the dot product expands to

$$\begin{aligned} \cos\theta &= \cos\delta_1 \cos\delta_2 \cos\alpha_1 \cos\alpha_2 + \cos\delta_1 \cos\delta_2 \sin\alpha_1 \sin\alpha_2 + \sin\delta_1 \sin\delta_2 \\ &= \sin\delta_1 \sin\delta_2 + \cos\delta_1 \cos\delta_2 \cos(\alpha_1 - \alpha_2), \end{aligned} \tag{2.10}$$

where the middle two terms were combined using $\cos\alpha_1 \cos\alpha_2 + \sin\alpha_1 \sin\alpha_2 = \cos(\alpha_1 - \alpha_2)$. Equation (2.10) is precisely the *spherical law of cosines*, and it is the fundamental tool for angular-distance problems on the celestial sphere or on the surface of the Earth (replacing $\delta \to \phi$, latitude, and $\alpha \to \lambda$, longitude).

Worked example (angular separation of Sirius and Regulus). Converting both stars' coordinates to decimal degrees ($1^{\mathrm{h}} = 15°$),

$$\text{Sirius: } (\alpha_1, \delta_1) \approx (101.287°, -16.716°), \qquad \text{Regulus: } (\alpha_2, \delta_2) \approx (152.093°, 11.967°),$$

Eq. (2.10) gives

$$\cos\theta = \sin(-16.716°)\sin(11.967°) + \cos(-16.716°)\cos(11.967°)\cos(50.806°),$$

yielding $\theta \approx 57.8°$.

Worked example (baseline chord between two ground stations). The identical formula, with $\delta \to \phi$ and $\alpha \to \lambda$, gives the central angle γ subtended at the Earth's center by two observatories:

$$\cos\gamma = \sin\phi_1 \sin\phi_2 + \cos\phi_1 \cos\phi_2 \cos(\lambda_2 - \lambda_1).$$

For GMRT $(19.09°\mathrm{N}, 74.05°\mathrm{E})$ and Ooty $(11.38°\mathrm{N}, 76.66°\mathrm{E})$ this angle, combined with the Earth's radius, gives the straight-line (chord) distance – worked out in Chapter 5 in the context of interferometry.

2.8 The ecliptic coordinate system

Definition 2.6 (Ecliptic coordinates) Ecliptic longitude λ *and* ecliptic latitude β *describe position relative to the plane of the Earth's orbit (the ecliptic plane) rather than the Earth's equatorial plane. λ is measured eastward from the vernal equinox direction within the ecliptic plane, and β is measured perpendicular to that plane.*

The corresponding unit vector is

$$\hat{\mathbf{r}}_{\mathrm{ecl}} = \begin{pmatrix} \cos\beta \cos\lambda \\ \cos\beta \sin\lambda \\ \sin\beta \end{pmatrix}.$$

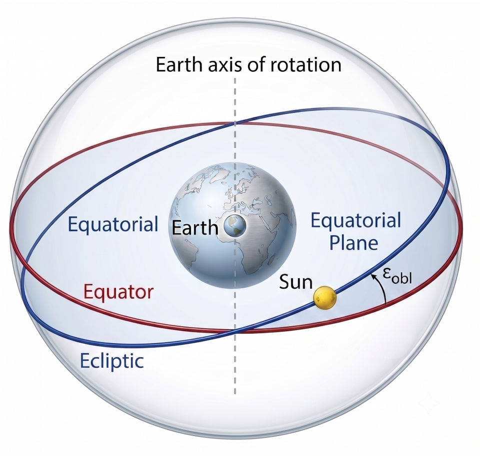


Figure 2.6: The obliquity of the ecliptic, $\epsilon \simeq 23.44°$, is the tilt of the Earth's rotation axis (and hence of the celestial equator) relative to the plane of the Earth's orbit (the ecliptic).

2.8.1 Deriving the equatorial–ecliptic rotation

The equatorial and ecliptic systems share the same fundamental reference direction: the vernal equinox $\hat{\gamma}$ is, by definition, the direction in which the two great circles (celestial equator and ecliptic) intersect, so both systems use this same direction as their x-axis. They differ only by a rotation about this common x-axis through the obliquity of the ecliptic, $\epsilon \simeq 23.44°$ – the tilt of the Earth's rotation axis relative to its orbital plane.

For a rotation of coordinate axes by angle θ about a common axis, a fixed vector's components in the new (primed) basis are obtained from its components in the old basis via the standard two-dimensional rotation matrix acting on the two orthogonal directions:

$$\begin{pmatrix} \hat{y}' \\ \hat{z}' \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} \hat{y} \\ \hat{z} \end{pmatrix}.$$

Equivalently, expressing the *old* basis vectors in terms of the new axes (which is what we need, since we want to re-express the equatorial unit vector $\hat{\mathbf{r}}_{\rm eq}$, built from $\hat{x}, \hat{y}, \hat{z}$, in the ecliptic frame), one inverts the rotation to find

$$\hat{y} = \begin{pmatrix} \cos\theta \\ -\sin\theta \end{pmatrix}, \qquad \hat{z} = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}$$

(components given in the primed, i.e. ecliptic, $\hat{y}', \hat{z}'$ basis). With $\theta = \epsilon$, the ecliptic components of the equatorial basis vectors are

$$\hat{x} \to \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad \hat{y} \to \begin{pmatrix} 0 \\ \cos\epsilon \\ -\sin\epsilon \end{pmatrix}, \qquad \hat{z} \to \begin{pmatrix} 0 \\ \sin\epsilon \\ \cos\epsilon \end{pmatrix}.$$

Writing $\hat{\mathbf{r}}_{\rm eq} = \hat{x}\cos\delta\cos\alpha + \hat{y}\cos\delta\sin\alpha + \hat{z}\sin\delta$ and substituting the ecliptic components of each basis vector,

$$\begin{aligned} \begin{pmatrix} \cos\beta\cos\lambda \\ \cos\beta\sin\lambda \\ \sin\beta \end{pmatrix} &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\epsilon & \sin\epsilon \\ 0 & -\sin\epsilon & \cos\epsilon \end{pmatrix} \begin{pmatrix} \cos\delta\cos\alpha \\ \cos\delta\sin\alpha \\ \sin\delta \end{pmatrix} \\ &= \begin{pmatrix} \cos\delta\cos\alpha \\ \cos\delta\sin\alpha\cos\epsilon + \sin\delta\sin\epsilon \\ -\cos\delta\sin\alpha\sin\epsilon + \sin\delta\cos\epsilon \end{pmatrix}. \end{aligned} \tag{2.11}$$

Reading off the third (latitude) component and using arctan of the ratio of the second to the first component for longitude (retaining the correct quadrant),

Result 2.4

$$\beta = \sin^{-1}\left(\sin\delta\cos\epsilon - \cos\delta\sin\alpha\sin\epsilon\right), \qquad \lambda = \tan^{-1}\left(\frac{\cos\delta\sin\alpha\cos\epsilon + \sin\delta\sin\epsilon}{\cos\delta\cos\alpha}\right).$$

Worked example. For Sirius, $\alpha \approx 101.287°, \delta \approx -16.716°$, one finds $\lambda \approx -75.9°$, $\beta \approx -39.6°$. For Regulus, $\alpha \approx 152.093°, \delta \approx 11.967°$, one finds $\lambda \approx -69.6°$, $\beta \approx 0.46°$. Regulus's small ecliptic latitude reflects the fact that it lies very close to the zodiacal band, along which the Sun, Moon, and planets travel.

Chapter 3

Astrometry: Distances, Motions, and the Space Velocity Vector

3.1 Trigonometric parallax

Definition 3.1 (Parallax) *As the Earth orbits the Sun, a nearby star appears to shift slightly against the background of much more distant stars. The* parallax ϖ *is half the total angular shift observed over the year, equal to the angle subtended at the star by the Earth-Sun distance (1 AU).*

By definition of the parsec as the distance at which 1 AU subtends exactly $1''$ of parallax,

Result 3.1

$$d[pc] = \frac{1}{\varpi['']}.$$

This follows directly from the small-angle relation $\varpi[\mathrm{rad}] = 1\,\mathrm{AU}/d$ together with the definitions of the parsec and the arcsecond; no further physics is required, only the geometric fact that a right triangle with a very small opposite angle ϖ and adjacent side d has opposite side $\approx d\varpi$ (in radians), here set equal to 1 AU.

Worked example. For Sirius, $\varpi = 41.13$ mas $= 0.04113''$, so

$$d = \frac{1}{0.04113} \simeq 24.31 \text{ pc}.$$

The same idea applies to resolved binary orbits: if a star's true orbital semi-major axis subtends an angular semi-major axis θ_i as seen from Earth, then since both ϖ and θ_i are angles subtended by lengths at the same distance d,

$$a_i = d\,\theta_i[\mathrm{rad}] = \frac{\theta_i}{\varpi}\,\mathrm{AU},$$

a convenient shortcut used later in binary-star problems that avoids ever computing d explicitly.

3.2 Cartesian position from parallax and coordinates

Once d, α, and δ are known, the star's three-dimensional position vector relative to the Sun follows immediately from the equatorial unit vector of Section 2.1, scaled by the distance:

$$\vec{r} = d\begin{pmatrix}\cos\delta\cos\alpha \\ \cos\delta\sin\alpha \\ \sin\delta\end{pmatrix}.$$

For Sirius ($d \simeq 24.31$ pc, $\alpha \simeq 101.287°$, $\delta \simeq -16.716°$), this evaluates to $\vec{r} \simeq (-4.56,\ 22.84,\ -6.99)$ pc.

3.3 Proper motion and transverse velocity

Definition 3.2 (Proper motion) Proper motion *is the star's true angular motion on the sky, over and above the annual parallactic wobble, arising from its actual transverse (sideways) velocity relative to the Sun. It is usually resolved into components along right ascension,* $\mu_\alpha \equiv \dot{\alpha}\cos\delta$ *(the* $\cos\delta$ *accounting for the convergence of hour circles toward the poles), and along declination,* $\mu_\delta \equiv \dot{\delta}$*.*

An angular rate of motion μ (arcsec/yr) at distance d (pc) corresponds to a true transverse speed

$$v_\perp = \mu[\text{rad/yr}] \times d[\text{AU}].$$

Converting units: 1 AU/yr, expressed in convenient physical units, equals

$$1\ \text{AU/yr} = \frac{1.496 \times 10^{11}\ \text{m}}{3.156 \times 10^{7}\ \text{s}} \simeq 4.74047\ \text{km/s},$$

using the Julian year of 365.25 days. Combined with 1 pc = 206265 AU and $1'' = 1/206265$ rad, one AU per year at one parsec per arcsecond per year of proper motion works out to exactly this same constant, giving the standard astrometric formula

Result 3.2

$$v_\perp[km/s] = 4.74047\ \mu[\text{arcsec}\,/yr]\ d[pc].$$

This constant, 4.74047 km/s, is one of the most-used numbers in stellar kinematics and is worth memorizing: it converts an angular rate at a known distance directly into a physical speed.

Worked example (Sirius). With $\mu_\alpha = -248.73$ mas/yr, $\mu_\delta = -5.59$ mas/yr, and $d = 24.31$ pc,

$$v_\alpha = 4.74047 \times (-0.24873) \times 24.31 \simeq -28.67\ \text{km/s}, \qquad v_\delta \simeq -0.64\ \text{km/s}.$$

3.4 The full space velocity vector

The star's velocity relative to the Sun has, in general, three independent components: one along the line of sight (the *radial velocity* v_r, measured spectroscopically via the Doppler shift) and two transverse components (v_α, v_δ, from proper motion as above). These three components are naturally expressed along a local orthonormal triad attached to the star's current direction: the radial direction $\hat{\mathbf{r}}$ (equal to $\hat{\mathbf{r}}_{\text{eq}}$ of Section 2.1) together with two tangent directions $\hat{\boldsymbol{\alpha}}$ (increasing right ascension) and $\hat{\boldsymbol{\delta}}$ (increasing declination), obtained formally as

$$\hat{\boldsymbol{\alpha}} = \frac{\partial\hat{\mathbf{r}}/\partial\alpha}{|\partial\hat{\mathbf{r}}/\partial\alpha|} = \begin{pmatrix} -\sin\alpha \\ \cos\alpha \\ 0 \end{pmatrix}, \qquad \hat{\boldsymbol{\delta}} = \frac{\partial\hat{\mathbf{r}}/\partial\delta}{|\partial\hat{\mathbf{r}}/\partial\delta|} = \begin{pmatrix} -\sin\delta\cos\alpha \\ -\sin\delta\sin\alpha \\ \cos\delta \end{pmatrix}.$$

(These are exactly the unit tangent vectors one obtains by differentiating $\hat{\mathbf{r}}_{\text{eq}}$ with respect to each spherical angle in turn and normalizing, which is the general recipe for building a local orthonormal frame on any curved coordinate surface.) The full space velocity is then

Result 3.3

$$\vec{v} = v_r\hat{\mathbf{r}} + v_\alpha\hat{\boldsymbol{\alpha}} + v_\delta\hat{\boldsymbol{\delta}}, \qquad |\vec{v}| = \sqrt{v_r^2 + v_\alpha^2 + v_\delta^2},$$

the final equality holding because $\hat{\mathbf{r}}, \hat{\boldsymbol{\alpha}}, \hat{\boldsymbol{\delta}}$ are mutually orthogonal unit vectors.

Worked example (Sirius). With $v_r = 4.39$ km/s, $v_\alpha \simeq -28.67$ km/s, $v_\delta \simeq -0.64$ km/s, one finds $\vec{v} \simeq (27.33,\ 9.55,\ -1.88)$ km/s in equatorial Cartesian components, with magnitude $|\vec{v}| \simeq 29.01$ km/s.

3.5 The Doppler shift and radial velocity

Definition 3.3 (Doppler shift) *The relativistic Doppler formula for a source receding at radial speed v_r relative to an observer gives an observed wavelength*

$$\lambda' = \lambda_0 \sqrt{\frac{1+\beta}{1-\beta}}, \qquad \beta \equiv v_r/c.$$

Expanding for $\beta \ll 1$ (as is true for essentially all stars in the Galaxy) using a first-order Taylor expansion,

$$\sqrt{\frac{1+\beta}{1-\beta}} \approx (1+\tfrac{1}{2}\beta)(1+\tfrac{1}{2}\beta) \approx 1+\beta,$$

so that

Result 3.4

$$\frac{\Delta\lambda}{\lambda_0} \equiv \frac{\lambda' - \lambda_0}{\lambda_0} \simeq \frac{v_r}{c}.$$

This non-relativistic Doppler formula is what is used whenever a radial velocity is quoted for a star; it is exact in the limit $v_r \ll c$ and is an excellent approximation for any ordinary stellar velocity (tens of km/s against $c = 3 \times 10^5$ km/s).

Worked example (Hα line of Sirius). With $\lambda_0 = 656.46$ nm and $v_r = 4.39$ km/s,

$$\Delta\lambda = 656.46 \times \frac{4.39}{299792} \simeq 0.0096 \text{ nm},$$

so the observed wavelength is redshifted (Sirius is receding) to $\lambda_{\rm obs} \simeq 656.470$ nm.

3.6 Beyond trigonometric parallax: the cosmic distance ladder

Trigonometric parallax is the only *direct*, purely geometric method of measuring astronomical distances, but it loses precision rapidly with distance – a parallax of $\varpi = 1$ mas (near the limit of ground-based astrometry before space missions such as Hipparcos and Gaia) already corresponds to $d = 1000$ pc, and the associated fractional distance error scales as σ_ϖ/ϖ, which grows linearly with d. To reach the distances of other galaxies, astronomers instead rely on *standard candles*: classes of object whose intrinsic luminosity can be inferred from some other, distance-independent observable, so that the distance modulus of Section 4.2 can be applied without first knowing d.

Definition 3.4 (Cosmic distance ladder) *The* cosmic distance ladder *is the sequence of overlapping distance methods, each calibrated using the previous, more local rung: trigonometric parallax calibrates the luminosities of certain nearby standard candles, whose luminosities then allow distances to be measured for more distant objects too far away for a measurable parallax, and so on outward.*

The best-known rung is the *Cepheid period-luminosity relation* (the “Leavitt Law,” after its discoverer Henrietta Swan Leavitt, who noticed in observations of variable stars in the Small

Magellanic Cloud that longer-period Cepheids are systematically brighter): Cepheid variables are pulsating giant stars whose absolute magnitude M is tightly correlated with their pulsation period P via a relation of the form $M = a\log_{10} P + b$, with a, b calibrated using nearby Cepheids whose distances are known from parallax (or from cluster membership, historically). Once calibrated, simply timing a distant Cepheid's pulsation period gives its absolute magnitude, and comparing this to the observed apparent magnitude via the distance modulus formula of Section 4.2 gives d, entirely independent of parallax. Because Cepheids are luminous supergiants, this method extends usefully distant, and further rungs (most importantly Type Ia supernovae, whose peak luminosities are calibrated against Cepheid distances in their host galaxies) extend the ladder to cosmological distances, ultimately underlying the calibration of the Hubble constant. The parallax and proper-motion techniques of this chapter are thus not merely one method among several: they are the geometric foundation on which the entire distance ladder, and hence essentially all extragalactic distance measurement, rests.

3.7 A practical note on measuring radial velocity

The radial velocities used throughout this chapter are, in practice, not read off a single spectral line by eye but are obtained by *cross-correlating* an observed stellar spectrum against a template spectrum (either a model or another star of known, usually zero, velocity): the whole observed spectrum is shifted by a trial velocity and compared to the template over a wide range of trial shifts, and the shift that maximizes the correlation is adopted as v_r. This averages over many spectral lines simultaneously, which is why modern radial velocities can be determined to precisions far better than the width of any single spectral feature (down to the m/s level in exoplanet-hunting spectrographs), even though the underlying formula relating a wavelength shift to velocity is exactly the simple, non-relativistic Doppler relation derived above.

Chapter 4

Stellar Photometry: Flux, Magnitudes, and Radii

4.1 Flux and the inverse-square law

Definition 4.1 (Flux) *The* flux F *received from a star is the power per unit area crossing a surface perpendicular to the line of sight at the observer's location.*

If a star radiates isotropically with total luminosity L (power emitted in all directions), then at distance d this power is spread uniformly over the surface of an imaginary sphere of area $4\pi d^2$ centered on the star, so

Result 4.1

$$F = \frac{L}{4\pi d^2}.$$

This is the photometric analogue of Gauss's law: the same total "flow" (luminosity) passes through every sphere centered on the source, so the flux density falls as the inverse square of the distance purely from the growth of the sphere's surface area, with no assumption about the emission mechanism.

Worked example (Aldebaran). With $\varpi = 47$ mas, $d = 1/0.047 \simeq 21.27$ pc $\simeq 6.56\times10^{17}$ m, and $L = 439L_\odot \simeq 1.68 \times 10^{29}$ W (using $L_{\mathrm{bol},\odot} = 3.828 \times 10^{26}$ W),

$$F = \frac{1.68 \times 10^{29}}{4\pi(6.56 \times 10^{17})^2} \simeq 3.10 \times 10^{-8}\ \mathrm{W\,m^{-2}}.$$

4.2 The magnitude system

Definition 4.2 (Apparent magnitude) *The apparent magnitude scale is a historical, logarithmic scale of brightness, defined so that a difference of exactly 5 magnitudes corresponds to a flux ratio of exactly 100 (Pogson's ratio), i.e. each magnitude step corresponds to a flux ratio of* $100^{1/5} \approx 2.512$*:*

$$F \propto 10^{-m/2.5} \quad \Longrightarrow \quad m_1 - m_2 = -2.5\log_{10}\left(\frac{F_1}{F_2}\right).$$

Fainter *objects have* larger *(more positive) magnitudes – the scale runs backward relative to physical intuition, a historical accident inherited from the ancient Greek classification of stars by "magnitude" of brightness rank.*

Definition 4.3 (Absolute magnitude and distance modulus) *The* absolute magnitude M *of an object is defined as the apparent magnitude it* would have *if placed at a standard distance of exactly* 10 *pc.*

Using the inverse-square law to relate the flux at the true distance d to the (hypothetical) flux at 10 pc,

$$m - M = -2.5\log_{10}\left(\frac{L/4\pi d^2}{L/4\pi(10\,\mathrm{pc})^2}\right) = -2.5\log_{10}\left(\frac{(10\,\mathrm{pc})^2}{d^2}\right) = 5\log_{10}\left(\frac{d}{10\,\mathrm{pc}}\right),$$

which simplifies to the standard *distance modulus*:

Result 4.2

$$m - M = 5\log_{10}(d[pc]) - 5.$$

Because L cancels entirely out of this relation, the distance modulus depends only on distance, never on the intrinsic properties of the source; it is a pure statement of geometric dilution. By the same inverse-square argument, for two objects at the *same* distance, their magnitude difference reflects only the ratio of their luminosities,

$$M_1 - M_2 = -2.5\log_{10}\left(\frac{L_1}{L_2}\right).$$

Here M (and correspondingly m) may refer either to *bolometric* magnitude (integrated over all wavelengths, tracking total luminosity) or to magnitude in a specific photometric band (e.g. visual magnitude V), in which case only the flux in that band is used.

Worked example (Aldebaran). Using the solar reference $M_{\mathrm{bol},\odot} = 4.74$,

$$M_{\mathrm{bol}} = 4.74 - 2.5\log_{10}(439) \simeq -1.87.$$

With $d \simeq 21.27$ pc,

$$m_{\mathrm{bol}} = M_{\mathrm{bol}} + 5\log_{10}(21.27) - 5 \simeq -0.23.$$

4.3 The Stefan-Boltzmann law and stellar radii

Definition 4.4 (Stefan-Boltzmann law) *A blackbody at effective temperature T_{eff} radiates a flux $\sigma T_{\mathrm{eff}}^4$ per unit surface area (where $\sigma = 5.67\times10^{-8}\ W m^{-2} K^{-4}$ is the Stefan-Boltzmann constant). For a star modeled as a sphere of radius R radiating as a blackbody, the total luminosity is*

$$L_{\mathrm{bol}} = 4\pi R^2 \sigma T_{\mathrm{eff}}^4,$$

the product of the emitted flux per unit area and the star's total surface area $4\pi R^2$.

Given a measured luminosity (from flux and distance, as above) and an independently determined effective temperature (from the star's spectrum or color), the radius follows immediately:

Result 4.3

$$R = \sqrt{\frac{L}{4\pi\sigma T_{\mathrm{eff}}^4}}.$$

Worked example (Aldebaran). With $L \simeq 1.68 \times 10^{29}$ W and $T_{\mathrm{eff}} = 3900$ K,

$$R = \sqrt{\frac{1.68\times10^{29}}{4\pi(5.67\times10^{-8})(3900)^4}} \simeq 3.19\times10^{10}\ \mathrm{m} \simeq 45.9\,R_\odot.$$

Aldebaran is thus a giant star, roughly forty-six times the Sun's radius, in line with its status as a red giant.

4.4 Angular diameter

For a star of physical radius R at distance $d \gg R$, the small-angle approximation ($\tan\theta \approx \theta$ for $\theta \ll 1$ rad) gives the *angular radius* $\theta \approx R/d$, and hence the angular diameter

Result 4.4

$$\Theta = 2\theta \approx \frac{2R}{d}.$$

Worked example (Aldebaran). With $R \simeq 3.19 \times 10^{10}$ m and $d \simeq 6.56 \times 10^{17}$ m,

$$\Theta \simeq \frac{2(3.19 \times 10^{10})}{6.56 \times 10^{17}} \simeq 9.71 \times 10^{-8} \text{ rad}.$$

Using 1 rad $= 206265 \times 10^3$ mas, this converts to $\Theta \simeq 20.0$ mas – a striking illustration of just how small even a giant star's disk appears from interstellar distances, and why resolving stellar disks directly requires interferometric techniques rather than simple imaging.

Chapter 5

Angular Resolution: Baselines, Diffraction, and Pixel Sampling

5.1 Chord distance between two points on the Earth

Given the latitude-longitude coordinates of two ground stations, the *central angle* γ they subtend at the Earth's center is obtained from the very same spherical law of cosines derived in Eq. (2.10), with $\delta \to \phi$ and $\alpha \to \lambda$:

$$\cos\gamma = \sin\phi_1 \sin\phi_2 + \cos\phi_1 \cos\phi_2 \cos(\lambda_2 - \lambda_1).$$

The straight-line (chord, not great-circle) distance between the two stations, needed for interferometric baselines, follows from elementary trigonometry of the isoceles triangle formed by the two radii R_E and the chord (Figure 5.1):

Result 5.1

$$C = 2R_E \sin\left(\frac{\gamma}{2}\right), \qquad R_E = 6371\ km.$$

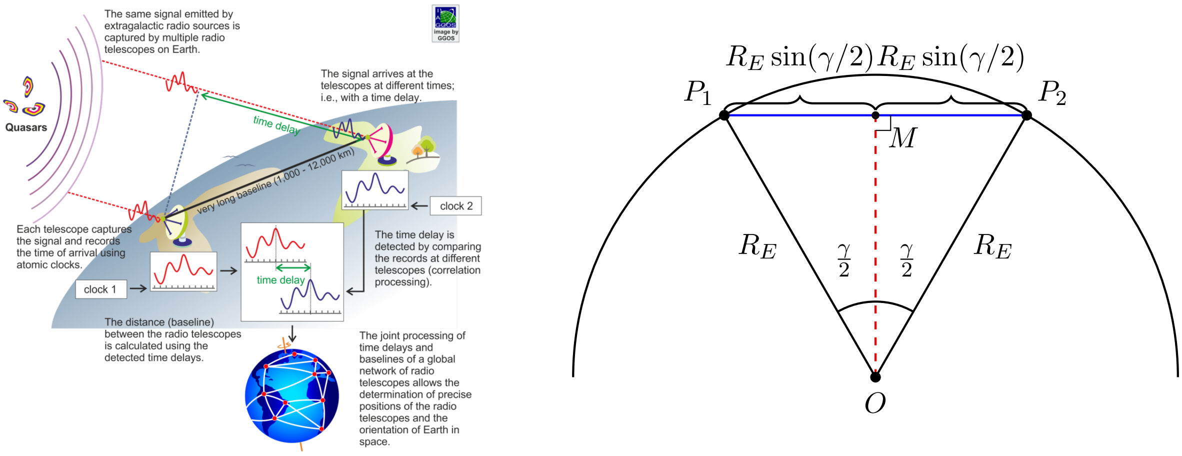


Figure 5.1: Very-long-baseline interferometry (VLBI) correlates signals received at widely separated radio telescopes, using the light-travel-time delay to synthesize an effective aperture equal to the physical separation of the stations. Geometrically, the straight-line baseline is the chord $C = 2R_E \sin(\gamma/2)$ subtended by the central angle γ between the two stations.

Worked example (GMRT and Ooty). With GMRT at $(19.09°\text{N}, 74.05°\text{E})$ and Ooty at $(11.38°\text{N}, 76.66°\text{E})$, the spherical law of cosines gives γ and hence $C \simeq 904.5$ km.

5.2 Interferometric angular resolution

Definition 5.1 (Interferometric resolution) *Two widely separated antennas observing the same source and correlating their signals behave, for the purposes of angular resolution, like the two opposite edges of a single enormous aperture of diameter equal to the baseline B. The angular resolution achievable is then approximately*

$$\theta \simeq \frac{\lambda}{B}.$$

This is the same functional form as ordinary diffraction-limited resolution for a filled aperture (see below), reflecting the fact that resolution is fundamentally set by the ratio of the observing wavelength to the largest dimension over which phase information about the incoming wavefront is combined – whether that dimension is a single mirror's diameter or the separation between two telescopes.

Worked example. At $f = 1000$ MHz, $\lambda = c/f = 0.30$ m. With $B \simeq 904.5$ km $= 9.045 \times 10^5$ m,

$$\theta \simeq \frac{0.30}{9.045 \times 10^5} \simeq 3.32 \times 10^{-7} \text{ rad} \simeq 68.4 \text{ mas},$$

illustrating how VLBI achieves milliarcsecond-scale resolution far beyond what any single dish could provide, by trading a filled aperture for a widely spaced, sparsely sampled one at the cost of a more complex image-reconstruction problem.

5.3 Telescope optics: focal length, plate scale, and pixel scale

Consider a two-mirror (Cassegrain) reflecting telescope, in which light collected by a large primary mirror of focal length F is intercepted by a smaller secondary mirror before reaching the actual focal plane behind the primary.

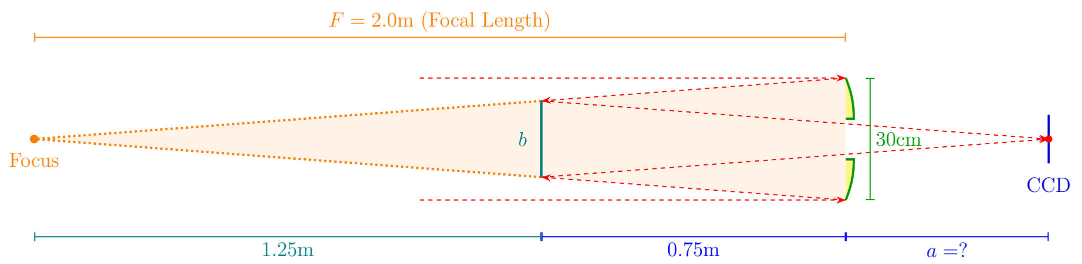


Figure 5.2: Cassegrain reflector telescope: light converging toward the primary focus is intercepted by the secondary mirror and refocused a distance a behind the primary onto the CCD.

Result 5.2 *[Similar triangles] If the secondary mirror sits a distance d_2 before the (would-be) primary focus, and the primary mirror has diameter D and focal length F, the secondary's diameter b satisfies*

$$\frac{b}{D} = \frac{F - d_2}{F},$$

because the converging cone of light forms similar triangles with apex at the primary focus.

Worked example. With $D = 30$ cm, $F = 2.0$ m, and the secondary 0.75 m from the primary (so $F - 0.75 = 1.25$ m remains to the would-be focus), $b = 30(1.25/2.0) = 18.75$ cm. If the CCD sits a further a behind the primary, with the fold accounted for by the secondary, $a = 1.25 - 0.75 = 0.50$ m.

Definition 5.2 (Plate scale and pixel scale) *The* plate scale *relates a small angle on the sky, $d\theta$, to the corresponding physical displacement ds in the focal plane:*

$$P \equiv \frac{d\theta}{ds} = \frac{\theta}{y},$$

using the small-angle relation $\tan\theta \approx \theta = y/F$ *(a source displaced by angle θ from the optical axis lands a distance $y = F\theta$ from the axis in the focal plane, by simple similar-triangle geometry of the converging beam, Figure 5.3). The* pixel scale *is simply the plate scale evaluated for a displacement equal to one physical pixel size p:*

$$\theta_{\text{pix}} = \frac{p}{F}.$$

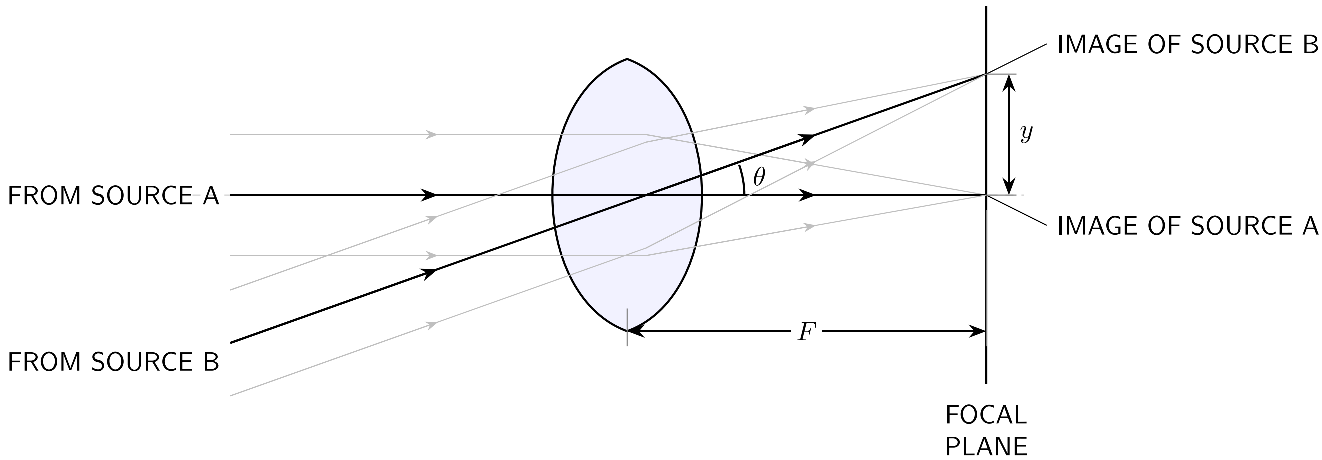


Figure 5.3: Two point sources separated by angle θ on the sky are imaged a physical distance $y = F\theta$ apart in the focal plane, by similar-triangle geometry of the chief rays through the lens (or mirror) center.

Worked example. With $p = 20$ μm and $F = 2$ m,

$$\theta_{\text{pix}} = \frac{20 \times 10^{-6}}{2} = 1.0 \times 10^{-5} \text{ rad} \simeq 2.06''/\text{pixel}.$$

5.4 Diffraction-limited resolution

Even a perfectly manufactured, aberration-free telescope cannot focus light from a point source to a mathematical point, because the finite aperture diameter D diffracts the incoming wavefront. The resulting image (the Airy pattern) has a bright central peak surrounded by faint rings; the Rayleigh criterion defines two point sources as "just resolved" when the central maximum of one source's diffraction pattern falls on the first minimum of the other's.

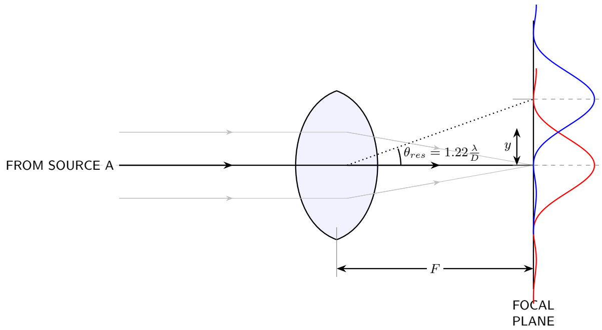


Figure 5.4: The Rayleigh criterion: two sources are marginally resolved when the central maximum of each Airy pattern coincides with the first minimum of the other, separated by the diffraction angle $\theta_1 = 1.22\lambda/D$.

Result 5.3 *[Rayleigh criterion] For a circular aperture of diameter D observing at wavelength λ, the diffraction-limited angular resolution is*

$$\theta_{\text{diff}} = 1.22\frac{\lambda}{D} \text{ rad.}$$

The factor 1.22 arises from the precise location of the first zero of the Airy diffraction pattern (the first zero of the Bessel function J_1 governing diffraction by a circular aperture) and is a fixed numerical constant, not an adjustable parameter.

Worked example. With $\lambda = 500$ nm and $D = 30$ cm,

$$\theta_{\text{diff}} = 1.22\frac{500 \times 10^{-9}}{0.30} \simeq 2.03 \times 10^{-6} \text{ rad} \simeq 0.42''.$$

5.5 Comparing pixel scale to the diffraction limit: Nyquist sampling

Definition 5.3 (Nyquist sampling criterion) *To avoid losing angular resolution to coarse pixelation (undersampling), the detector's pixel scale should be no coarser than about half the diffraction-limited resolution:*

$$\textit{Pixel Scale} \leq \frac{\theta_{\text{diff}}}{2}.$$

This is the astronomical instance of the general Nyquist-Shannon sampling principle: to faithfully represent a signal (here, the spatial variation of brightness across a diffraction-blurred point-spread function) without aliasing, one must sample at least twice per resolution element.

Worked example. With a pixel scale of $2.06''$/pixel against a diffraction limit of $0.42''$,

$$\frac{\text{Pixel Scale}}{\theta_{\text{diff}}} = \frac{2.06}{0.42} \simeq 4.9.$$

Since this ratio is far above the Nyquist requirement of $1/2$, a single pixel here spans nearly five diffraction-limited resolution elements: the optical system's diffraction-limited resolving power is being wasted by coarse sampling, an example of an undersampled, poorly matched instrument configuration. Improving this configuration would require either a smaller pixel size or a longer effective focal length (larger plate scale in the opposite sense, i.e. finer angular sampling per pixel).

Chapter 6

The Two-Body Problem

6.0.1 Central force problem

A central force acts along the line joining two particles and depends only on their separation $r = |\vec{x}_1 - \vec{x}_2|$. Such a force is a gradient of a scalar potential $V(r)$:

$$\vec{F}(\vec{x}_1, \vec{x}_2) = \nabla V(r).$$

Newton's equations for the pair are

$$\frac{d^2}{dt^2}\begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix}\begin{bmatrix} \vec{x}_1 \\ \vec{x}_2 \end{bmatrix} = \begin{bmatrix} -\nabla V(r) \\ \nabla V(r) \end{bmatrix}. \tag{6.1}$$

Six second-order ODEs $\Rightarrow$ 12 constants of integration in the general solution.

6.0.2 Separating centre-of-mass motion

With $M = m_1 + m_2$, $\mu = m_1 m_2/M$, transform to relative and centre-of-mass coordinates,

$$\begin{bmatrix} \vec{r} \\ \vec{X} \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ m_1/M & m_2/M \end{bmatrix}\begin{bmatrix} \vec{x}_1 \\ \vec{x}_2 \end{bmatrix}, \qquad \begin{bmatrix} \vec{x}_1 \\ \vec{x}_2 \end{bmatrix} = \begin{bmatrix} m_2/M & 1 \\ -m_1/M & 1 \end{bmatrix}\begin{bmatrix} \vec{r} \\ \vec{X} \end{bmatrix}.$$

Substituting into (6.1) and left-multiplying by $\begin{bmatrix} m_2/M & -m_1/M \\ 1 & 1 \end{bmatrix}$ diagonalizes the mass matrix and decouples the two sectors:

$$\mu\,\ddot{\vec{r}} = -\nabla V(r), \tag{6.2}$$

$$M\,\ddot{\vec{X}} = 0. \tag{6.3}$$

Equation (6.3) integrates trivially: $\vec{X}(t) = (\vec{P}/M)t + \vec{X}_0$, with $\vec{P}, \vec{X}_0$ six constants fixing the inertial frame in which the barycentre is at rest at the origin. Everything of dynamical interest is in the *relative* coordinate $\vec{r}$, governed by (6.2) — a one-body central-force problem with reduced mass μ.

6.0.3 Conserved quantities

Angular momentum. Define $\vec{L} = \vec{r} \times \vec{p} = \mu\,\vec{r} \times \dot{\vec{r}}$. Then, because we are considering central force $\ddot{\vec{r}} \propto \hat{r}$ and hence,

$$\dot{\vec{L}} = \mu\big[\dot{\vec{r}} \times \dot{\vec{r}} + \vec{r} \times \ddot{\vec{r}}\big] = 0.$$

$\vec{L}$ supplies three constants of motion: its magnitude L, and two angles fixing its direction — the **inclination** ι and **longitude of ascending node** Ω (Euler angles of the orbital plane relative to a reference plane). Conservation of $\vec{L}$ confines the motion to the plane $\perp \vec{L}$.

Energy. $E = K + V$ with $K = \frac{1}{2}\mu\dot{\vec{r}}^2$ (the centre-of-mass kinetic term is a constant and drops out). Then

$$\dot{E} = \mu\dot{\vec{r}}\cdot\ddot{\vec{r}} + \dot{V} = \mu\dot{\vec{r}}\cdot\ddot{\vec{r}} + \nabla V\cdot\dot{\vec{r}} = \big(\mu\ddot{\vec{r}} + \nabla V\big)\cdot\dot{\vec{r}} = 0$$

by (6.2). So E is conserved for *any* central potential $V(r)$, not just gravity. Therefore, **total energy and angular momentum** constitutes four constant of integration. We will learn that there is one more, known as Runge Lenz vector which points towards periapsis and hence encodes the orientation of periapsis and the last constant of integration which is needed specifies the initial phase of the relative orbit (t_p)

$$\underbrace{\mathbf{R}_0, \mathbf{V}_{\rm CM}}_{6} + \underbrace{a, e, i, \Omega, \omega, t_p}_{6} = 12.$$

6.1 Coordinate Systems and Orbital Elements

Fix an equatorial (reference) frame with $\hat{\Upsilon}$ (vernal equinox / first point of Aries) along X and celestial north $\hat{N}$ along Z; $\hat{Y}$ completes the right-handed triad. For a source at $(\mathrm{RA}, \mathrm{DEC}) = (\alpha, \delta)$ the line of sight is

$$\hat{R} = \cos\delta\cos\alpha\,\hat{\Upsilon} + \cos\delta\sin\alpha\,\hat{Y} + \sin\delta\,\hat{N},$$

with in-sky-plane axes (pointing along increasing RA, DEC)

$$\hat{\alpha} = \frac{\frac{\partial\hat{R}}{\partial\alpha}}{\left|\frac{\partial\hat{R}}{\partial\alpha}\right|} = -\sin\alpha\,\hat{\Upsilon} + \cos\alpha\,\hat{Y}, \qquad \hat{\delta} = \frac{\frac{\partial\hat{R}}{\partial\delta}}{\left|\frac{\partial\hat{R}}{\partial\delta}\right|} = -\sin\delta\cos\alpha\,\hat{\Upsilon} - \sin\delta\sin\alpha\,\hat{Y} + \cos\delta\,\hat{N}.$$

The orbital plane has its own frame: origin at the barycentre, X-axis along the **line of nodes** (through the ascending node and the descending node), Z-axis along $\vec{L}$. Denote the unit vector towards the ascending node by $\hat{\Omega}$. Then

$$\begin{pmatrix}\hat{\Omega}\\ \hat{y}\\ \hat{L}\end{pmatrix} = \begin{pmatrix}1 & 0 & 0\\ 0 & \cos\iota & \sin\iota\\ 0 & -\sin\iota & \cos\iota\end{pmatrix}\begin{pmatrix}\cos\Omega & \sin\Omega & 0\\ -\sin\Omega & \cos\Omega & 0\\ 0 & 0 & 1\end{pmatrix}\begin{pmatrix}\hat{\alpha}\\ \hat{\delta}\\ \hat{R}\end{pmatrix}. \tag{6.4}$$

Hence,

$$\hat{\Omega} = \cos\Omega\,\hat{\alpha} + \sin\Omega\,\hat{\delta}, \tag{6.5}$$

$$\hat{y} = -\sin\Omega\cos\iota\,\hat{\alpha} + \cos\Omega\cos\iota\,\hat{\delta} + \sin\iota\,\hat{R}, \tag{6.6}$$

$$\hat{L} = \sin\Omega\sin\iota\,\hat{\alpha} - \cos\Omega\sin\iota\,\hat{\delta} + \cos\iota\,\hat{R}, \tag{6.7}$$

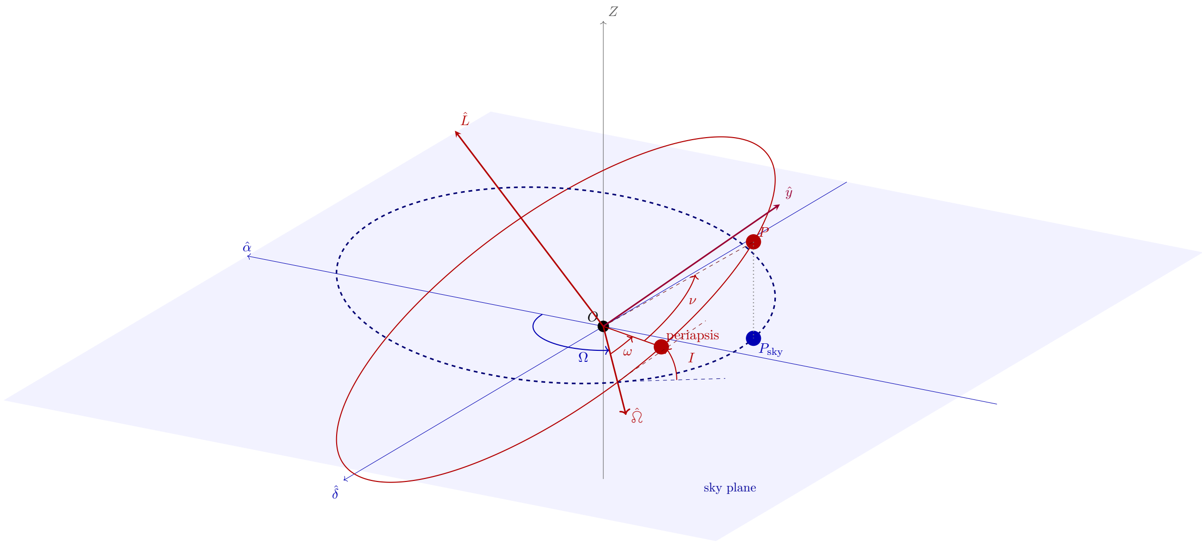


Figure 6.1: Orientation of an orbit relative to sky plane. Ω (longitude of ascending node) is measured from the reference direction α to the ascending node along the sky plane; ω (argument of periapsis) is measured from the ascending node to perigee *within* the orbital plane; ι (inclination) is the dihedral angle between the two planes, read off at the node; ν (true anomaly) is measured from perigee to the heavenly body's current position. This is exactly (Ω, ι, ω) of Eqs. (6.5)–(6.6) describe.

so that $\iota = \arccos\left(\hat{L} \cdot \hat{R}\right)$ is the angle between orbital and reference planes, and $(\hat{\Omega}, \hat{y}, \hat{L})$ is a right-handed orthonormal triad spanning the orbital plane plus its normal. In this plane use polar coordinates (r, ϕ) measured from $\hat{\Omega}$:

$$\hat{r} = \cos\phi\,\hat{\Omega} + \sin\phi\,\hat{y}, \qquad \hat{\phi} = -\sin\phi\,\hat{\Omega} + \cos\phi\,\hat{y}. \tag{6.8}$$

The projected orbit in the sky plane : If the object in its orbital plane as the coordinate $(r\cos(\omega+\nu), r\sin(\omega+\nu))$, the corresponding coordinate in the sky plane could be found as:

$$\begin{pmatrix} X \\ Y \\ Z \end{pmatrix}_{\text{Sky Plane}} = \begin{pmatrix} \cos\Omega & -\sin\Omega & 0 \\ \sin\Omega & \cos\Omega & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos i & -\sin i \\ 0 & \sin i & \cos i \end{pmatrix} \begin{pmatrix} r\cos(\omega+\nu) \\ r\sin(\omega+\nu) \\ 0 \end{pmatrix}_{\text{Orbital Plane}} . \tag{6.9}$$

Then,

$$X = r\left(\cos(\omega+\nu)\cos\Omega - \sin(\omega+\nu)\sin\Omega\cos i\right), \tag{6.10}$$

$$Y = r\left(\cos(\omega+\nu)\sin\Omega + \sin(\omega+\nu)\cos\Omega\cos i\right). \tag{6.11}$$

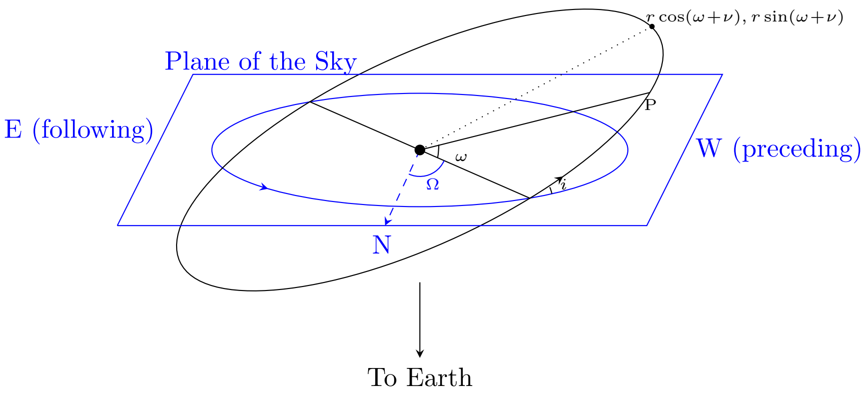


Figure 6.2: In this North becomes the local X axis and West becomes the Y axis of the sky plane.

One can set $\Omega = 0$ by another set of basis in the sky plane and then the analysis simplifies immensely

$$X = r\cos(\omega + \nu), \tag{6.12}$$
$$Y = r\sin(\omega + \nu)\cos i. \tag{6.13}$$

We find that the semi minor axis shrinks by the factor of $\cos i$.

The six orbital elements. As the orbit solution unfolds below, six constants emerge in total: Ω, ι (orientation of the plane, from $\vec{L}$), a, e (size and shape, from E, L), ω (orientation of the ellipse *within* the plane, the **argument of periapsis**), and T_0 (time of periapsis passage, fixing where on the ellipse the body is at a given time). Figure 6.1 shows all of these together.

6.2 Planar Motion and Kepler's Second Law

Areal velocity. The area swept by $\vec{r}$ in time dt is $dA = \int_0^{2\pi}\int_0^{r(\phi)} r dr d\phi = \frac{1}{2}\int_0^{2\pi} r^2(\phi) d\phi$, so

$$\frac{dA}{dt} = \frac{r^2\dot{\phi}}{2} = \frac{L}{2\mu} = \text{const.} \tag{6.14}$$

This is **Kepler's second law**: equal areas in equal times. It follows purely from $\vec{L}$ being conserved, i.e. from the force being central — no dependence on the specific form of $V(r)$.

Equations of motion in (r, ϕ)**.** With $Z \parallel \vec{L}$ and $X \parallel \hat{\Omega}$, $\vec{r} = r\hat{r}$, and

$$\dot{\vec{r}} = \dot{r}\,\hat{r} + r\dot{\hat{r}} = \dot{r}\,\hat{r} + r\dot{\phi}\,\hat{\phi}, \qquad \ddot{\vec{r}} = \left(\ddot{r} - r\dot{\phi}^2\right)\hat{r} + \left(2\dot{r}\dot{\phi} + r\ddot{\phi}\right)\hat{\phi}.$$

where we used $\nabla_\mu \mathbf{e}_\nu = \Gamma^\alpha_{\mu\nu}\mathbf{e}_\alpha$ which gives $\frac{\mathrm{d}\hat{r}}{\mathrm{d}t} = \frac{\partial \hat{r}}{\partial \phi}\dot{\phi} = \dot{\phi}\Gamma^\mu_{\phi r}\mathbf{e}_\mu = \dot{\phi}\frac{\mathbf{e}_\phi}{r} = \dot{\phi}\hat{\phi}$. Equation (6.2) splits into radial and tangential parts:

$$\mu\left(\ddot{r} - r\dot{\phi}^2\right) = -\nabla V(r), \tag{6.15}$$
$$\mu\left(2\dot{r}\dot{\phi} + r\ddot{\phi}\right) = 0 \implies \frac{\mathrm{d}(r^2\dot{\phi})}{\mathrm{d}t} = 0. \tag{6.16}$$

This was supposed to be a coupled differential equation. But because of equation (6.16) is just angular momentum conservation, i.e. Kepler's second law again, since $\vec{L} = \mu\vec{r}\times\dot{\vec{r}} = \mu r^2\dot{\phi}\,\hat{L}$. So

$$\dot{\phi} = \frac{L}{\mu r^2}, \tag{6.17}$$

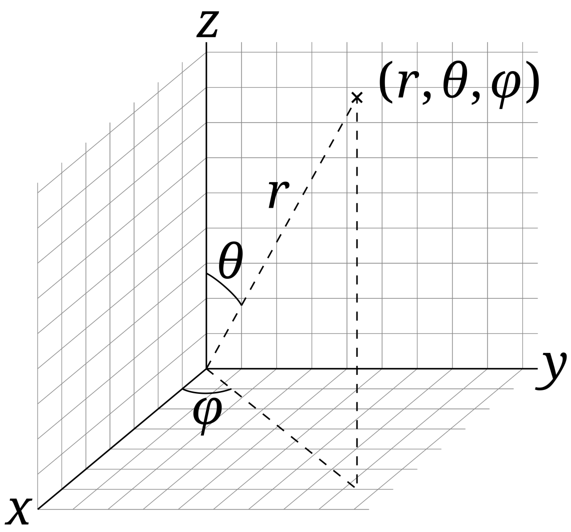


Figure 6.3: Schematic description of spherical polar coordinates.

and (6.15) becomes, eliminating $\dot{\phi}$,

$$\mu\ddot{r} - \frac{L^2}{\mu r^3} = -\nabla V(r). \tag{6.18}$$

6.3 The Orbit Equation

To get the orbit *shape* $r(\phi)$ we change independent variable from t to ϕ using (6.17): $\frac{dr}{dt} = \dot{\phi}\frac{dr}{d\phi} = \frac{L}{\mu r^2}\frac{dr}{d\phi}$, and

$$\frac{d^2r}{dt^2} = \frac{L}{\mu r^2}\frac{d}{d\phi}\left[\frac{L}{\mu r^2}\frac{dr}{d\phi}\right] = \frac{L^2}{\mu^2 r^4}\left[\frac{d^2r}{d\phi^2} - \frac{2}{r}\left(\frac{dr}{d\phi}\right)^2\right].$$

Substituting into (6.18) and dividing by $L^2/(\mu r^4)$:

$$\frac{1}{r^4}\left[\frac{d^2r}{d\phi^2} - \frac{2}{r}\left(\frac{dr}{d\phi}\right)^2\right] - \frac{1}{r^3} = -\frac{\mu}{L^2}\nabla V(r).$$

The standard trick is the **Binet substitution** $r = 1/\rho$: then $dr/d\phi = -\rho^{-2}d\rho/d\phi$ and $d^2r/d\phi^2 = -\rho^{-2}d^2\rho/d\phi^2 + 2\rho^{-3}(d\rho/d\phi)^2$. Substituting and simplifying (the $(d\rho/d\phi)^2$ terms cancel), one obtains the **orbit equation**:

$$\boxed{\frac{d^2\rho}{d\phi^2} + \rho = \frac{\mu}{L^2}\frac{1}{\rho^2}\nabla V\left(\frac{1}{\rho}\right).} \tag{6.19}$$

This is a purely geometric ODE for the orbit's shape, valid for any central potential.

6.4 Keplerian Orbits

For Newtonian gravity, $V(r) = -GM\mu/r$, so $\nabla V(r) = GM\mu\rho^2\,\hat{r}$. Equation (6.19) becomes

$$\frac{d^2\rho}{d\phi^2} + \rho = \frac{GM\mu^2}{L^2}.$$

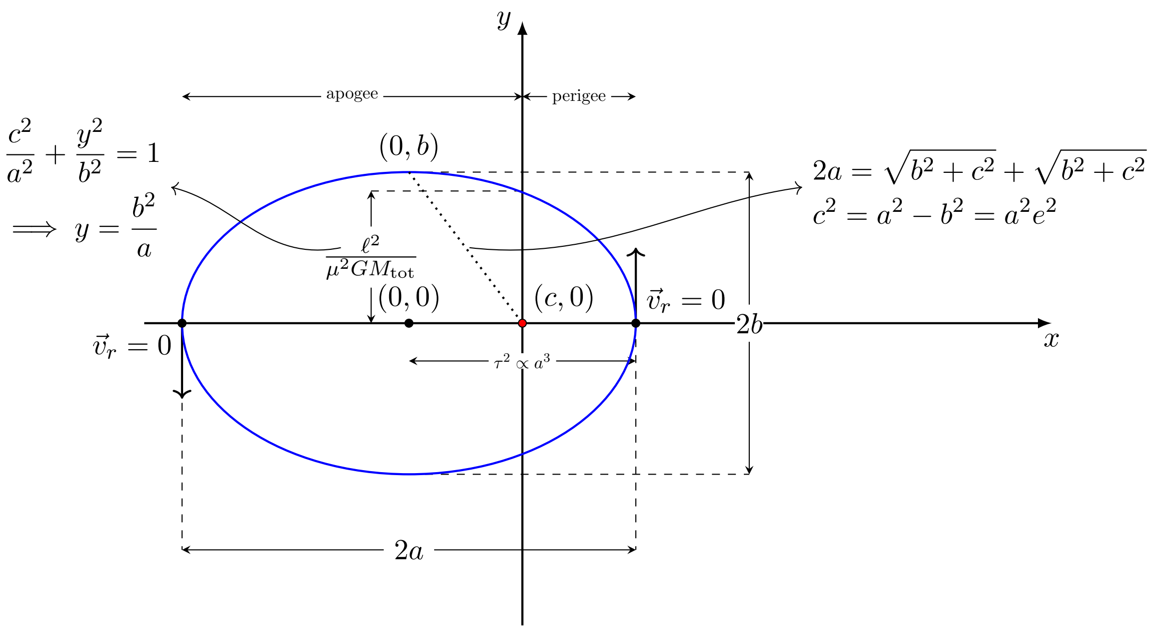
Shifting $\tilde{\rho} = \rho - GM\mu^2/L^2$ turns this into the simple-harmonic-oscillator equation $\ddot{\tilde{\rho}} + \tilde{\rho} = 0$ (derivatives in ϕ), with solution $\tilde{\rho} = A\cos(\phi - \omega)$ for constants A, ω. In terms of $r = 1/\rho$:

$$r(\phi) = \left(\frac{GM\mu^2}{L^2} + A\cos(\phi - \omega)\right)^{-1} = \frac{L^2/(GM\mu^2)}{1 + \frac{L^2 A}{GM\mu^2}\cos(\phi - \omega)}.$$

Defining the **semi-latus rectum** $\lambda \equiv L^2/(GM\mu^2)$ as the value of r when $\phi = \omega + 90°$ and **eccentricity** $e \equiv \lambda A$,

$$\boxed{r(\phi) = \frac{\lambda}{1 + e\cos(\phi - \omega)}.} \tag{6.20}$$

This is the polar equation of a conic section with one focus at the origin, semi-latus rectum λ, eccentricity e, and major axis rotated by ω (the **argument of periapsis** — periapsis occurs at $\phi = \omega$).



Two more shape parameters follow:

$$a = \frac{\lambda}{1 - e^2} \text{ (semi-major axis)}, \qquad b = a\sqrt{1 - e^2} \text{ (semi-minor axis)}.$$

One can check this directly from the equation of the ellipse,

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \tag{6.21}$$

by writing the polar coordinates about the focus as

$$x = ae + r(\phi)\cos\phi, \tag{6.22}$$
$$y = r(\phi)\sin\phi. \tag{6.23}$$

Substitution gives

$$\frac{(ae + r\cos\phi)^2}{a^2} + \frac{r^2\sin^2\phi}{a^2(1 - e^2)} = 1. \tag{6.24}$$

Solving for r then yields

$$r(\phi) = \frac{a(1 - e^2)}{1 + e\cos\phi}. \tag{6.25}$$

Eccentricity	Conic section
$e = 0$	circle
$0 < e < 1$	ellipse ($E < 0$, bound)
$e = 1$	parabola ($E = 0$)
$e > 1$	hyperbola ($E > 0$, unbound)

Bound, closed elliptical orbits ($E < 0$) constitute **Kepler's first law**. In the original (unreduced) frame,

$$\vec{x}_1 = \frac{m_2}{M}\vec{r}, \qquad \vec{x}_2 = -\frac{m_1}{M}\vec{r}.$$

6.5 Energy and Angular Momentum in Terms of a, e

6.5.1 Energy

$$E = \frac{1}{2}\mu\dot{\vec{r}}^2 - \frac{GM\mu}{r} = \frac{1}{2}\mu\big(\dot{r}^2 + r^2\dot{\phi}^2\big) - \frac{GM\mu}{r} = \frac{1}{2}\mu\dot{r}^2 + \frac{L^2}{2\mu r^2} - \frac{GM\mu}{r}.$$

Since E is conserved, evaluate at periapsis ($\phi = \omega$), where $r = a - ae = \lambda/(1+e)$ and $\dot{r} = 0$ (turning point of r):

$$E = 0 + \frac{L^2(1+e)^2}{2\mu\lambda^2} - \frac{GM\mu(1+e)}{\lambda} = \frac{GM\mu^2\lambda(1+e)^2}{2\mu\lambda^2} - \frac{GM\mu(1+e)}{\lambda} = -\frac{GM\mu}{2\lambda}(1-e^2),$$

using $L^2 = GM\mu^2\lambda$ (below). With $a = \lambda/(1-e^2)$,

$$\boxed{E = -\frac{GM\mu}{2a}.} \tag{6.26}$$

Since $\lambda > 0$ always, the sign of E tracks e: bound orbits ($E < 0$) have $e < 1$, $a > 0$; unbound orbits ($E \geq 0$) have $e \geq 1$; hyperbolic orbits formally have $a < 0$.

6.5.2 Angular momentum

Both at apoapsis ($r = a + ae = \frac{\lambda}{1-e}$) and periapsis($r = a - ae = \frac{\lambda}{1+e}$), we have $\dot{r} = 0$,

$$\frac{L^2(1+e)^2}{2\mu\lambda^2} - \frac{GM\mu(1+e)}{\lambda} = \frac{L^2(1-e)^2}{2\mu\lambda^2} - \frac{GM\mu(1-e)}{\lambda} \tag{6.27}$$

Solving for L:

$$L = \sqrt{GM\mu^2\lambda}, \qquad \text{i.e.} \qquad L^2 = GM\mu^2 a(1-e^2), \tag{6.28}$$

and, using (6.7),

$$\vec{L} = \sqrt{GM\mu^2\lambda}\,\big(-\sin\Omega\sin\iota\,\hat{\alpha} + \cos\Omega\sin\iota\,\hat{\delta} + \cos\iota\,\hat{R}\big).$$

6.5.3 Instantaneous kinetic energy

From $E = K + V$: $K = \frac{1}{2}\mu\dot{\vec{r}}^2 = E - V = GM\mu\left(\frac{1}{r} - \frac{1}{2a}\right)$ — the *vis-viva* equation. Splitting between the two bodies via $\vec{x}_i$: $K_1 = \frac{1}{2}m_1 v_1^2 = \frac{m_2}{M}K$, $K_2 = \frac{m_1}{M}K$.

Inverting for a, e. Equations (6.26)–(6.28) give the orbital elements directly from the (conserved) energy and angular momentum:

$$a = -\frac{GM\mu}{2E}, \qquad e = \sqrt{1 + \frac{2EL^2}{(GM)^2\mu^3}}. \tag{6.29}$$

6.6 True and Eccentric Anomaly; Kepler's Equation

We now want $r(t)$ (and hence $\phi(t)$), not just $r(\phi)$. From the vis-viva relation,

$$\dot{r} = \sqrt{\frac{2}{\mu}\left(E - \frac{L^2}{2\mu r^2} + \frac{GM\mu}{r}\right)} = \sqrt{GM}\,\frac{1}{r}\left(2r - \frac{r^2}{a} - \lambda\right)^{1/2},$$

using (6.26)–(6.28) to trade E, L for a, λ. Then

$$\int_{t_0}^{t} dt = \frac{1}{\sqrt{GM}} \int_{r_0}^{r} \frac{r'\,dr'}{(2r' - r'^2/a - \lambda)^{1/2}} \tag{6.30}$$

$$t - t_0 = \sqrt{\frac{a}{GM}} \int_{r_0}^{r} \frac{r'dr'}{\sqrt{a^2e^2 - (r'-a)^2}} \qquad \text{(using } \lambda = a(1-e^2)\text{)}$$

6.6.1 Elliptical orbits: eccentric anomaly

For $a > 0$, substitute

$$r = a\big(1 - e\cos u\big), \tag{6.31}$$

where u is the **eccentric anomaly** (this choice guarantees $r > 0$ throughout, since $-1 \leq \cos u \leq 1$). Then $dr = ae\sin u\,du$, and the bracket under the square root simplifies to $ae^2\sin^2 u$, so the square root is simply $\sqrt{a}\,e\,|\sin u|$. The integral collapses to

$$t - t_0 = \frac{a^{3/2}}{\sqrt{GM}} \int_0^u (1 - e\cos u')\,du' = \frac{a^{3/2}}{\sqrt{GM}}\big[u - e\sin u\big],$$

where t_0 (**epoch**) is the time at which $u = 0$ — the sixth and final constant of integration, T_0.

Relating true anomaly v and eccentric anomaly u: Define the **true anomaly** $v \equiv \phi - \omega$ (so $r = \lambda/(1 + e\cos v)$ from (6.20)). Equating this with (6.31):

$$\frac{1 - e\cos u}{1 - e^2} = \frac{1}{1 + e\cos v} \quad \Longrightarrow \quad \cos v = \frac{\cos u - e}{1 - e\cos u}. \tag{65a}$$

Consistently,

$$\sin v = \frac{\sqrt{1-e^2}\sin u}{1 - e\cos u}, \qquad \tan v = \frac{\sqrt{1-e^2}\sin u}{\cos u - e}, \qquad \tan\frac{v}{2} = \sqrt{\frac{1+e}{1-e}}\tan\frac{u}{2}. \tag{6.32}$$

At $u = 0$ and $u = \pi$, $v = u$ — periapsis and apoapsis coincide for the two anomalies, as they must by symmetry.

6.6.2 Hyperbolic trajectories

For $a < 0$ (and $e > 1$), the analogous substitution is $r = a(1 - e\cosh u)$ (using $\cosh u \geq 1$ keeps $r > 0$ since $a < 0$). Carrying through the same integral gives the **hyperbolic** anomaly relation

$$t - t_0 = \frac{(-a)^{3/2}}{\sqrt{GM}}\big[e\sinh u - u\big],$$

with true-anomaly relations

$$\cos v = \frac{e - \cosh u}{e\cosh u - 1}, \qquad \sin v = \frac{\sqrt{e^2-1}\sinh u}{e\cosh u - 1}, \qquad \tan\frac{v}{2} = \sqrt{\frac{e+1}{e-1}}\tanh\frac{u}{2}. \tag{6.33}$$

6.7 Kepler's Third Law

For one complete orbit, u advances by 2π while t advances by the orbital period $2\pi/n$ ($n \equiv$ mean motion):

$$\frac{2\pi}{n} = \frac{a^{3/2}}{\sqrt{GM}}\big[2\pi - e\sin(2\pi)\big] = \frac{a^{3/2}}{\sqrt{GM}} \cdot 2\pi \quad \Longrightarrow \quad \boxed{n = \sqrt{\frac{GM}{a^3}}, \qquad n^2a^3 = GM.} \tag{6.34}$$

6.8 Kepler's Equation

Substituting $n = \sqrt{GM/a^3}$ into $t - t_0 = (a^{3/2}/\sqrt{GM})(u - e\sin u)$ and defining the **mean anomaly** $\ell \equiv n(t - t_0)$:

$$\boxed{\ell = u - e\sin u} \qquad \text{(elliptical Kepler's equation).} \tag{6.35}$$

This transcendental equation must be solved numerically (or by series, §6.11) for $u(\ell)$, i.e. $u(t)$. The hyperbolic analogue (with $n \equiv GM/(-a)^{3/2}$) is

$$\ell = e\sinh u - u. \tag{6.36}$$

6.9 Keplerian Parametrization

6.9.1 Elliptical orbits

Collecting results:

$$r = a(1 - e\cos u), \qquad \phi = v + \omega, \tag{6.37}$$

$$\tan\frac{v}{2} = \sqrt{\frac{1-e}{1+e}}\tan\frac{u}{2}, \qquad \ell = u - e\sin u. \tag{6.38}$$

Differentiating Kepler's equation (6.35) in t: $n = \dot{u}(1 - e\cos u)$, so

$$\dot{u} = \frac{n}{1 - e\cos u}. \tag{6.39}$$

Differentiating $r = a(1 - e\cos u)$:

$$\dot{r} = ae\sin u\,\dot{u} = \frac{an\,e\sin u}{1 - e\cos u}. \tag{6.40}$$

Differentiating the v-u relation and using (6.32), one finds $dv/du = \sqrt{1-e^2}/(1 - e\cos u)$, so

$$\dot{\phi} = \dot{v} = \frac{dv}{du}\dot{u} = \frac{n\sqrt{1-e^2}}{(1 - e\cos u)^2}. \tag{6.41}$$

6.9.2 Hyperbolic trajectories

Analogously, with $r = -a(e\cosh u - 1)$, $\phi = v + \omega$, $\tan(v/2) = \sqrt{(e+1)/(e-1)}\tanh(u/2)$, $\ell = e\sinh u - u$:

$$\dot{u} = \frac{n}{e\cosh u - 1}, \qquad \dot{r} = -an\frac{e\sinh u}{e\cosh u - 1}, \qquad \dot{\phi} = \dot{v} = \frac{n\sqrt{e^2-1}}{(e\cosh u - 1)^2}. \tag{6.42}$$

6.10 Hyperbolic-Trajectory Parameters

For unbound ($e > 1$) encounters, four derived quantities are standard:

Hyperbolic excess velocity v_∞: from $\frac{1}{2}\mu\dot{\vec{r}}_\infty^2 = -GM\mu/(2a)$ (recall $a < 0$),

$$|\dot{\vec{r}}_\infty| = \sqrt{-\frac{GM}{a}}. \tag{6.43}$$

Scattering angle θ_∞: half the angle between the two asymptotes,

$$\cos\theta_\infty = -\lim_{u\to\infty}\frac{e-\cosh u}{e\cosh u - 1} = \frac{1}{e}. \tag{6.44}$$

Periapsis distance:

$$r_p = \min_v \frac{a(1-e^2)}{1+e\cos v} = -a(e-1). \tag{6.45}$$

Semi-minor axis / impact parameter: the semi-minor axis $b = -a\sqrt{e^2-1}$ satisfies $-ab = \lambda$. The impact parameter b' (miss distance of the unperturbed asymptote) is

$$b' = (-a + r_p)\sin\theta_\infty = -ae\sin\theta_\infty = -a\sqrt{e^2-1} = b,$$

i.e. impact parameter and semi-minor axis coincide.

Flight-path angle (angle between $\dot{\vec{r}}$ and the direction $\perp \hat{r}$ in-plane):

$$\sin\varphi = \frac{\dot{\vec{r}}\cdot\hat{r}}{|\dot{\vec{r}}|} = \frac{\dot{r}}{\sqrt{\dot{r}^2 + r^2\dot{\phi}^2}}. \tag{6.46}$$

6.11 Fourier Solution of Kepler's Equation

Kepler's equation $u - \ell = e\sin u$ can be solved as a Fourier series in ℓ:

$$e\sin u = \sum_{s=1}^{\infty} A_s \sin(s\ell).$$

Using $A_s = \frac{2}{\pi}\int_0^\pi (u-\ell)\sin(s\ell)\,d\ell$, integrating by parts (the boundary term vanishes since $u = \ell$ at $\ell = 0, \pi$), and substituting $d\ell = (1 - e\cos u)\,du$ (from Kepler's equation) turns the ℓ-integral into a u-integral:

$$A_s = \frac{2}{s\pi}\int_0^\pi \cos(su - se\sin u)\,du = \frac{2}{s}J_s(se),$$

recognizing the integral representation of the Bessel function of the first kind, $J_s(x) = \frac{1}{\pi}\int_0^\pi \cos(s\tau - x\sin\tau)\,d\tau$. So

$$u(\ell) = \ell + \sum_{s=1}^{\infty}\frac{2}{s}J_s(se)\sin(s\ell). \tag{6.47}$$

This is the classical Bessel-function series solution to Kepler's equation.

6.12 The Laplace–Runge–Lenz Vector

Define $\vec{A} \equiv \vec{p} \times \vec{L} - GM\mu^2 \hat{r}$ (with $\vec{p} = \mu\dot{\vec{r}}$). For the pure Kepler problem,

$$\dot{\vec{A}} = \dot{\vec{p}} \times \vec{L} - GM\mu^2 \dot{\hat{r}} = -\frac{GM\mu}{r^2}\big(\hat{r} \times \vec{L}\big) - GM\mu^2 \dot{\phi}\,\hat{\phi}.$$

Using $\vec{L} = L\hat{L}$, planar motion identities $\hat{r} \times \hat{L} = -\hat{\phi}$, and $\dot{\phi} = L/(\mu r^2)$ from (6.17):

$$\dot{\vec{A}} = -GM\mu\left(\frac{L}{r^2}\hat{\phi} - \mu\dot{\phi}\,\hat{\phi}\right) = 0,$$

so $\vec{A}$ is conserved for the unperturbed problem — it is a *fourth*, non-obvious constant of the Kepler problem (beyond $E, \vec{L}$), reflecting the extra "hidden" $SO(4)$ symmetry of the $1/r$ potential.

Direction and magnitude of $\vec{A}$. From the definition it is clear that $\vec{A}$ is planar. Therefore, we can decompose it as $\vec{A} = A_r\hat{r} + A_\phi\hat{\phi}$.

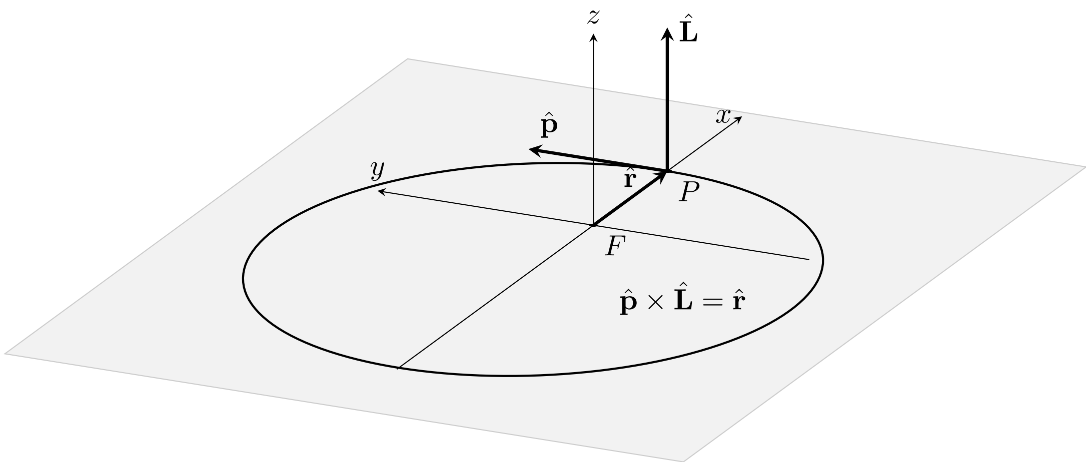


Figure 6.4: The Runge Lenz vector points radially outward at perigee. Since it is conserved, the direction and magnitude stays fixed during the entire evolution.

At periapsis the velocity is purely tangential ($\vec{p} \perp \hat{r}$), so $\hat{r} \times \vec{A}\big|_{\text{peri}} = \hat{r} \times (\vec{p} \times \vec{L})\big|_{\text{peri}} = \big[\vec{p}(\hat{r} \cdot \vec{L}) - \vec{L}(\hat{r} \cdot \vec{p})\big]_{\text{peri}} = 0$ (both dot products vanish there). Also $\hat{r} \cdot \vec{A}|_{\text{peri}} = A$ (from the relation above with $v = 0$ at periapsis). Together these mean $\hat{A} = \hat{r}|_{\text{peri}}$: $\vec{A}$ points from the focus *towards periapsis.*

Since $\vec{A}$ points radially outwards at the Periapsis. We find the magnitude by doting $\vec{A}$ into $\vec{r}$:

$$\vec{A} \cdot \vec{r} = (\vec{p} \times \vec{L}) \cdot \vec{r} - GM\mu^2 r \tag{6.48}$$
$$= p_i L_j \epsilon_{ijk} r_k - GM\mu^2 r = p_i L_j \epsilon_{kij} r_k - GM\mu^2 r \tag{6.49}$$
$$= \underbrace{(\vec{r} \times \vec{p})}_{\vec{L}} \cdot \vec{L} - GM\mu^2 r = L^2 - GM\mu^2 r, \tag{6.50}$$

i.e., writing $\vec{A} \cdot \vec{r} = Ar\cos v$ (with v the angle between $\vec{A}$ and $\vec{r}$), using

$$r = \frac{L^2/(GM\mu^2)}{1 + e\cos v} \implies GM\mu^2 r = \frac{L^2}{1 + e\cos v}.$$

We find:

$$Ar\cos v = L^2 - GM\mu^2 r, \tag{6.51}$$

$$A\frac{L^2/GM\mu^2}{1+e\cos v}\cos v = L^2\left(1 - \frac{1}{1+e\cos v}\right) = \frac{L^2 e\cos v}{1+e\cos v} \tag{6.52}$$

Hence

$$\boxed{A = GM\mu^2 e.} \tag{6.53}$$

Using (6.8):

$$\hat{A} = \cos\omega\,\hat{\Omega} + \sin\omega\,\hat{y}. \tag{6.54}$$

Equivalently, the **eccentricity vector** $\vec{e} \equiv \vec{A}/(GM\mu^2) = e\hat{A} = e(\cos\omega, \sin\omega, 0)$ is a conserved vector encoding (e, ω) together. Since e, Ω, ι are already determined by $(E, \vec{L})$, the only *new* information in $\vec{A}$ is ω, i.e. the in-plane orientation of the ellipse or the direction of periapsis from the line of ascending node.

6.13 Perturbed Orbits

We now allow an additional force $\vec{F}$ in the relative equation of motion:

$$\boxed{\mu\ddot{\vec{r}} = -\frac{GM\mu}{r^2}\hat{r} + \vec{F}.} \tag{6.55}$$

The central Newtonian term defines the Kepler problem; $\vec{F}$ is the perturbing force and need not be central.

The momentum is

$$\vec{p} = \mu\dot{\vec{r}}, \tag{6.56}$$

and, in the orbital plane,

$$\vec{p} = \mu\dot{r}\,\hat{r} + \frac{L}{r}\,\hat{\phi}, \qquad L = \mu r^2\dot{\theta}. \tag{6.57}$$

The purpose of orbital perturbation theory is to represent the actual trajectory by Keplerian orbital elements which vary with time, and to derive their rates directly from (6.55).

6.13.1 Osculating Kepler orbit

For $\vec{F} = 0$, the Kepler solution can be written in terms of six constants

$$Q = (Q_1, \ldots, Q_6) = (a, e, \iota, \Omega, \omega, T_0), \tag{6.58}$$

so that

$$\vec{r} = \vec{r}(Q, t), \qquad \vec{p} = \vec{p}(Q, t). \tag{6.59}$$

For fixed Q, these functions satisfy the unperturbed equations of motion.

For the perturbed problem, let

$$Q_i = Q_i(t). \tag{6.60}$$

The actual trajectory is represented in terms of a set of time-dependent osculating orbital elements $Q(t)$ as

$$\vec{r}(t) = \vec{r}_{\rm osc}\big(Q(t), t\big), \qquad \vec{p}(t) = \vec{p}_{\rm osc}\big(Q(t), t\big). \tag{6.61}$$

Here, for fixed values of Q, the functions $\vec{r}_{\rm osc}(Q, t)$ and $\vec{p}_{\rm osc}(Q, t)$ describe the corresponding Keplerian orbit. The time dependence of $Q(t)$ accounts for the slow evolution of this osculating Keplerian orbit under the perturbing force.

At each instant, the osculating Kepler orbit is chosen to have the same position and momentum as the actual trajectory. Thus,

$$\vec{r}_{\text{osc}}\big(Q(t),t\big)=\vec{r}(t), \qquad \vec{p}_{\text{osc}}\big(Q(t),t\big)=\vec{p}(t). \tag{6.62}$$

Importantly, this representation does not imply that the actual trajectory is itself Keplerian. Rather, it is represented at each instant by the Keplerian orbit that osculates it at that instant.

The corresponding accelerations are different:

$$\dot{\vec{p}}_{\text{actual}}=-\frac{GM\mu}{r^2}\hat{r}+\vec{F}, \tag{6.63}$$

$$\dot{\vec{p}}_{\text{osc}}=-\frac{GM\mu}{r^2}\hat{r}. \tag{6.64}$$

Hence

$$\boxed{\dot{\vec{p}}_{\text{actual}}-\dot{\vec{p}}_{\text{osc}}=\vec{F}.} \tag{6.65}$$

The two trajectories therefore agree in position and momentum at the instant at which the osculating orbit is constructed, but they have different subsequent evolution.

6.13.2 Variation of parameters

Differentiating (6.61),

$$\dot{\vec{r}}=\left.\frac{\partial\vec{r}}{\partial t}\right|_Q+\sum_{i=1}^{6}\frac{\partial\vec{r}}{\partial Q_i}\dot{Q}_i. \tag{6.66}$$

For fixed Q, the first term is the velocity of the Kepler solution. The osculating condition requires this Kepler velocity to coincide with the actual velocity:

$$\dot{\vec{r}}=\left.\frac{\partial\vec{r}}{\partial t}\right|_Q. \tag{6.67}$$

Therefore

$$\boxed{\sum_{i=1}^{6}\frac{\partial\vec{r}}{\partial Q_i}\dot{Q}_i=0.} \tag{6.68}$$

This is a vector equation and therefore supplies three scalar conditions.

Now differentiate $\vec{p}(Q(t),t)$:

$$\dot{\vec{p}}=\left.\frac{\partial\vec{p}}{\partial t}\right|_Q+\sum_{i=1}^{6}\frac{\partial\vec{p}}{\partial Q_i}\dot{Q}_i. \tag{6.69}$$

For fixed Q, $\vec{p}(Q,t)$ obeys the Kepler equation,

$$\left.\frac{\partial\vec{p}}{\partial t}\right|_Q=-\frac{GM\mu}{r^2}\hat{r}. \tag{6.70}$$

The actual momentum obeys

$$\dot{\vec{p}}=-\frac{GM\mu}{r^2}\hat{r}+\vec{F}. \tag{6.71}$$

Subtracting the Kepler contribution gives

$$\boxed{\sum_{i=1}^{6}\frac{\partial\vec{p}}{\partial Q_i}\dot{Q}_i=\vec{F}.} \tag{6.72}$$

Equations (6.68) and (6.72) form six scalar equations for the six quantities $\dot{Q}_i$. This is the formal variation-of-parameters construction of the osculating elements.

The perturbing force therefore enters through the change in the instantaneous phase-space state; the Kepler orbit is redefined at each instant from that state.

6.13.3 Perturbed Kepler invariants

The Keplerian energy, angular momentum and eccentricity vector are defined from the instantaneous state by

$$E \equiv \frac{p^2}{2\mu} - \frac{GM\mu}{r}, \tag{6.73}$$

$$\vec{L} \equiv \vec{r} \times \vec{p}, \tag{6.74}$$

$$\vec{A} \equiv \frac{\vec{p} \times \vec{L}}{\mu} - GM\mu\,\hat{r}. \tag{6.75}$$

For the unperturbed Kepler problem these are constants. In the perturbed problem they become time-dependent quantities.

Their magnitudes determine the instantaneous Keplerian parameters:

$$E = -\frac{GM\mu}{2a(t)}, \tag{6.76}$$

$$L^2 = GM\mu^2 a(t)(1 - e^2(t)), \tag{6.77}$$

$$A \equiv |\vec{A}| = GM\mu^2 e(t). \tag{6.78}$$

Thus a and e may be regarded as functions of the actual state even when the full trajectory is not Keplerian.

6.13.4 Exact Rate of the Energy

Differentiate (6.73):

$$\dot{E} = \frac{\vec{p}}{\mu} \cdot \dot{\vec{p}} + \frac{GM\mu}{r^2}\dot{r}. \tag{6.79}$$

Using

$$\dot{\vec{p}} = -\frac{GM\mu}{r^2}\hat{r} + \vec{F} \tag{6.80}$$

and

$$\frac{\vec{p}}{\mu} \cdot \hat{r} = \dot{r}, \tag{6.81}$$

we obtain

$$\dot{E} = \dot{\vec{r}} \cdot \left(-\frac{GM\mu}{r^2}\hat{r} + \vec{F}\right) + \frac{GM\mu}{r^2}\dot{r} \tag{6.82}$$

$$= -\frac{GM\mu}{r^2}\dot{r} + \dot{\vec{r}} \cdot \vec{F} + \frac{GM\mu}{r^2}\dot{r} \tag{6.83}$$

$$= \boxed{\dot{E} = \dot{\vec{r}} \cdot \vec{F}.} \tag{6.84}$$

6.13.5 Exact Rate of angular momentum

From

$$\vec{L} = \vec{r} \times \vec{p}, \tag{6.85}$$

$$\dot{\vec{L}} = \dot{\vec{r}} \times \vec{p} + \vec{r} \times \dot{\vec{p}}. \tag{6.86}$$

Since $\vec{p} = \mu\dot{\vec{r}}$,

$$\dot{\vec{r}} \times \vec{p} = 0, \tag{6.87}$$

and hence

$$\dot{\vec{L}} = \vec{r} \times \left(-\frac{GM\mu}{r^2}\hat{r} + \vec{F} \right). \tag{6.88}$$

The central term vanishes because $\vec{r} \parallel \hat{r}$, giving

$$\boxed{\dot{\vec{L}} = \vec{r} \times \vec{F}.} \tag{6.89}$$

Thus only the component of $\vec{F}$ perpendicular to $\vec{r}$ produces torque.

6.13.6 Exact Rate of the Runge-Lenz vector

Differentiate

$$\vec{A} = \vec{p} \times \vec{L} - GM\mu^2\,\hat{r} : \tag{6.90}$$

$$\dot{\vec{A}} = \dot{\vec{p}} \times \vec{L} + \vec{p} \times \dot{\vec{L}} - GM\mu^2\,\dot{\hat{r}}. \tag{6.91}$$

Insert

$$\dot{\vec{p}} = -\frac{GM\mu}{r^2}\hat{r} + \vec{F} \tag{6.92}$$

and

$$\dot{\vec{L}} = \vec{r} \times \vec{F}. \tag{6.93}$$

Then

$$\dot{\vec{A}} = -\frac{GM\mu}{r^2}\hat{r} \times \vec{L} + \vec{F} \times \vec{L} + \vec{p} \times (\vec{r} \times \vec{F}) - GM\mu^2\,\dot{\hat{r}}. \tag{6.94}$$

For the Kepler part,

$$\dot{\hat{r}} = \frac{\vec{L} \times \vec{r}}{\mu r^3}, \tag{6.95}$$

so

$$-GM\mu^2\,\dot{\hat{r}} = -\frac{GM\mu}{r^2}\,\left(\vec{L} \times \hat{r}\right) \tag{6.96}$$

$$= \frac{GM\mu}{r^2}\,\hat{r} \times \vec{L}. \tag{6.97}$$

The two central terms cancel. Therefore

$$\boxed{\dot{\vec{A}} = \vec{F} \times \vec{L} + \vec{p} \times (\vec{r} \times \vec{F}).} \tag{6.98}$$

This equation is the starting point for the perturbative evolution of the eccentricity and the orientation of periapsis.

6.14 Planar perturbations

We now specialize to a perturbing force confined to the orbital plane:

$$\boxed{\vec{F} = F_r \hat{r} + F_\theta \hat{\phi}.} \tag{6.99}$$

The angular momentum vector is perpendicular to the plane,

$$\vec{L} = L\hat{L}. \tag{6.100}$$

Consequently the plane itself remains fixed:

$$\boxed{\dot{i} = 0, \qquad \dot{\Omega} = 0.} \tag{6.101}$$

In the orbital plane,

$$\vec{p} = \mu\dot{r}\,\hat{r} + \mu r\dot{\theta}\,\hat{\phi} = \mu\dot{r}\,\hat{r} + \frac{L}{r}\hat{\phi}. \tag{6.102}$$

The equations of motion follow directly from (6.55):

$$\mu(\ddot{r} - r\dot{\theta}^2) = -\frac{GM\mu}{r^2} + F_r, \tag{6.103}$$

$$\mu(r\ddot{\theta} + 2\dot{r}\dot{\theta}) = F_\theta. \tag{6.104}$$

The second equation gives

$$\frac{dL}{dt} = \frac{d}{dt}(\mu r^2\dot{\theta}) = rF_\theta, \tag{6.105}$$

so

$$\boxed{\dot{L} = rF_\theta.} \tag{6.106}$$

The energy equation becomes

$$\dot{E} = \dot{r}F_r + r\dot{\theta}F_\theta \tag{6.107}$$

$$= \boxed{\dot{r}F_r + \frac{L}{\mu r}F_\theta.} \tag{6.108}$$

6.14.1 Keplerian geometry of the instantaneous ellipse

Define the true anomaly

$$f \equiv \theta - \omega. \tag{6.109}$$

For the instantaneous Kepler orbit,

$$\boxed{r = \frac{\lambda}{1 + e\cos f},} \tag{6.110}$$

where

$$\boxed{\lambda \equiv a(1 - e^2) = \frac{L^2}{GM\mu^2}.} \tag{6.111}$$

The instantaneous Keplerian angular velocity is therefore

$$\dot{\theta} = \frac{L}{\mu r^2}. \tag{6.112}$$

We can obtain the instantaneous radial velocity by starting from the general time-dependent orbital elements,

$$a(t) = a_0 + \delta a(t), \qquad e(t) = e_0 + \delta e(t), \qquad \omega(t) = \omega_0 + \delta\omega(t), \tag{6.113}$$

where the variations δa, δe, and $\delta\omega \sim \mathcal{O}(\epsilon)$ are produced by the perturbing force. Thus, in general,

$$\dot{r} = \frac{\partial r}{\partial t} + \frac{\partial r}{\partial a}\dot{a} + \frac{\partial r}{\partial e}\dot{e} + \frac{\partial r}{\partial f}\dot{f}. \tag{6.114}$$

For the instantaneous osculating Keplerian orbit, we take the zeroth-order Keplerian orbit specified by the instantaneous values of the orbital elements and evaluate the radial velocity with these elements held fixed. Equivalently, we temporarily set the perturbation to zero while retaining the instantaneous state $(\vec{r}, \vec{p})$. Hence only the motion through the true anomaly contributes to the radial velocity, giving

$$\boxed{\dot{r} = \frac{dr}{df}\dot{f} = \frac{\lambda e \sin f}{(1 + e\cos f)^2}\frac{L}{\mu r^2} = \frac{Le\sin f}{\mu\lambda}.} \tag{6.115}$$

Since

$$\frac{\lambda}{r} = 1 + e\cos f, \tag{6.116}$$

we also have

$$\boxed{\frac{\mu r\dot{r}}{L} = \frac{e\sin f}{1 + e\cos f}.} \tag{6.117}$$

Define

$$\boxed{\xi \equiv \frac{e\sin f}{1 + e\cos f}.} \tag{6.118}$$

6.14.2 Rate of the semi-major axis

At a given instant, we associate the actual orbit with a Keplerian orbit having the same instantaneous energy. Denoting its semi-major axis by $a(t)$, its Keplerian energy is

$$E\Big|_{\text{osculating}} = -\frac{GM\mu}{2a}, \qquad \therefore \qquad \dot{E}\Big|_{\text{osculating}} = \frac{GM\mu}{2a^2}\dot{a}. \tag{6.119}$$

Since the perturbing force is of order ϵ,

$$F_r, F_\theta = O(\epsilon),$$

the rates of the orbital elements are themselves $O(\epsilon)$. Consequently, to obtain the leading, first-order evolution, the remaining quantities multiplying F_r and F_θ on the right-hand side of (6.108) need only be evaluated at zeroth order, i.e. using the unperturbed Keplerian orbit:

$$\dot{E}\Big|_{\text{osculating}} = \left(\dot{r}F_r + \frac{L}{\mu r}F_\theta\right)_0 \tag{6.120}$$

$$= \frac{L}{\mu\lambda}e\sin f\, F_r + \frac{L}{\mu r}F_\theta + O(\epsilon^2). \tag{6.121}$$

Hence we find:

$$\dot{a} = \frac{2a^2 L}{GM\mu^2}\left[\frac{e\sin f}{\lambda}F_r + \frac{1}{r}F_\theta\right]. \tag{6.122}$$

Since

$$L = \mu\sqrt{GM\lambda}, \qquad \lambda = a(1-e^2), \tag{6.123}$$

this becomes

$$\boxed{\dot{a} = \sqrt{\frac{4a^3}{GM\mu^2(1-e^2)}}\left[e\sin f\, F_r + (1+e\cos f)F_\theta\right].} \tag{6.124}$$

6.14.3 Rate of the eccentricity

Since

$$A = GM\mu^2 e, \tag{6.125}$$

it is sufficient to calculate $\dot{A}$ from (6.98). For the planar perturbation, we use

$$\vec{L} = L\hat{L} \tag{6.126}$$

together with

$$\vec{F}\times\vec{L} = L\left(F_r\hat{r} + F_\theta\hat{\phi}\right)\times\hat{L} \tag{6.127}$$

$$= L\left(-F_r\hat{\phi} + F_\theta\hat{r}\right). \tag{6.128}$$

$$\vec{p}\times(\vec{r}\times\vec{F}) = \left(\mu\dot{r},\hat{r} + \frac{L}{r}\hat{\phi}\right)\times\left(rF_\theta\hat{L}\right) \tag{6.129}$$

$$= LF_\theta\hat{r} - \mu r\dot{r}F_\theta\hat{\phi}. \tag{6.130}$$

Hence

$$\dot{\vec{A}} = \vec{F}\times\vec{L} + \vec{p}\times(\vec{r}\times\vec{F}) = 2LF_\theta\hat{r} - \left(LF_r + \mu r\dot{r}F_\theta\right)\hat{\phi}. \tag{6.131}$$

Using (6.118),

$$\boxed{\dot{\vec{A}} = L\left[2F_\theta\hat{r} - (F_r + \xi F_\theta)\hat{\phi}\right].} \tag{6.132}$$

For radial perturbation $\dot{\vec{A}} = -F_r L\hat{\phi}$. The unit vector in the direction of $\vec{A}$ is the direction of periapsis. Since the true anomaly is measured from periapsis,

$$\boxed{\hat{A} = \cos f\hat{r} - \sin f\hat{\phi}.} \tag{6.133}$$

Notice,

$$\dot{\vec{A}} = \dot{A}\hat{A} + A\dot{\hat{A}} \tag{6.134}$$

$$\dot{\vec{A}}\cdot\hat{A} = \dot{A} \qquad \left(\text{using } \frac{\mathrm{d}\hat{A}\cdot\hat{A}}{\mathrm{d}t} = 0\right)$$

Therefore

$$\dot{A} = \dot{\vec{A}}\cdot\hat{A} \tag{6.135}$$

$$= L\left[2F_\theta\cos f + (F_r + \xi F_\theta)\sin f\right]. \tag{6.136}$$

Thus

$$\boxed{\dot{A} = L\left[F_r\sin f + F_\theta\left(2\cos f + \xi\sin f\right)\right].} \tag{6.137}$$

Up to this point, all relations have been exact. We now evaluate the exact rate equation using the instantaneous osculating Keplerian orbit, replacing the corresponding quantities on both the left- and right-hand sides by their osculating values.

$$\dot{A}\Big|_{\text{osculating}} = L\left[F_r \sin f + F_\theta \left(2\cos f + \xi \sin f\right)\right]\Big|_{\text{osculating}} \tag{6.138}$$

$$GM\mu^2 \dot{e} = \sqrt{GM\mu^2 a(1-e^2)}\left[F_r \sin f + F_\theta \left(2\cos f + \xi \sin f\right)\right] \tag{6.139}$$

we obtain

$$\boxed{\dot{e} = \sqrt{\frac{a(1-e^2)}{GM\mu^2}}\left[F_r \sin f + F_\theta \left(2\cos f + \xi \sin f\right)\right].} \tag{6.140}$$

6.14.4 Rate of periapsis

Define the unit vector perpendicular to $\hat{A}$ in the direction of increasing true anomaly:

$$\boxed{\hat{q} = \sin f\,\hat{r} + \cos f\,\hat{\phi}.} \tag{6.141}$$

Differentiating (6.133) gives

$$\dot{\hat{A}} = -\dot{f}\sin f\,\hat{r} + \cos f\,\dot{\hat{r}} - \dot{f}\cos f\,\hat{\phi} - \sin f\,\dot{\hat{\phi}}, \tag{6.142}$$

with

$$\dot{\hat{r}} = \dot{\theta}\,\hat{\phi}, \qquad \dot{\hat{\phi}} = -\dot{\theta}\,\hat{r}. \tag{6.143}$$

Thus

$$\dot{\hat{A}} = (\dot{\theta} - \dot{f})\left(\sin f\,\hat{r} + \cos f\,\hat{\phi}\right) \tag{6.144}$$

$$= \dot{\omega}\,\hat{q}, \tag{6.145}$$

because

$$f = \theta - \omega \quad \Longrightarrow \quad \dot{f} = \dot{\theta} - \dot{\omega}. \tag{6.146}$$

Since

$$\dot{\vec{A}} = \dot{A}\,\hat{A} + A\dot{\hat{A}}, \tag{6.147}$$

projection onto $\hat{q}$ gives

$$A\dot{\omega} = \dot{\vec{A}}\cdot\hat{q}. \tag{6.148}$$

Using (6.132) and (6.141),

$$A\dot{\omega}\Big|_{\text{osculating}} = L\left[2F_\theta \sin f - (F_r + \xi F_\theta)\cos f\right]\Big|_{\text{osculating}}. \tag{6.149}$$

Using $A = GM\mu^2 e$, we find:

$$\boxed{\dot{\omega} = -\frac{L}{GM\mu^2 e}\left[F_r \cos f + F_\theta \left(\xi \cos f - 2\sin f\right)\right].} \tag{6.150}$$

Using

$$\frac{L}{GM\mu^2 e} = \frac{1}{e}\sqrt{\frac{a(1-e^2)}{GM\mu^2}}, \tag{6.151}$$

we obtain

$$\boxed{\dot{\omega} = -\frac{1}{e}\sqrt{\frac{a(1-e^2)}{GM\mu^2}}\left[F_r \cos f + F_\theta \left(\xi \cos f - 2\sin f\right)\right].} \tag{6.152}$$

6.14.5 Rate of the true anomaly

Since

$$f = \theta - \omega, \tag{6.153}$$

we have

$$\dot{f} = \dot{\theta} - \dot{\omega}. \tag{6.154}$$

The instantaneous Keplerian value of $\dot{\theta}$ is

$$\dot{\theta} = \frac{L}{\mu r^2}. \tag{6.155}$$

Now we apply the same osculating orbit idea and find:

$$\dot{f} = \frac{L}{\mu r^2} - \dot{\omega}\bigg|_{\text{osculating}} \tag{6.156}$$

$$= \sqrt{\frac{GM}{a^3}} \frac{(1 + e\cos f)^2}{(1-e^2)^{3/2}} - \dot{\omega} \tag{6.157}$$

where we used

$$r = \frac{a(1-e^2)}{1 + e\cos f} \tag{6.158}$$

and

$$L = \mu\sqrt{GMa(1-e^2)}. \tag{6.159}$$

6.14.6 Planar Gauss equations

Collecting the results,

$$\dot{a} = \sqrt{\frac{4a^3}{GM\mu^2(1-e^2)}} \left[e \sin f\, F_r + (1 + e\cos f)F_\theta\right], \tag{6.160}$$

$$\dot{e} = \sqrt{\frac{a(1-e^2)}{GM\mu^2}} \left[F_r \sin f + F_\theta \left(2\cos f + \xi \sin f\right)\right], \tag{6.161}$$

$$\dot{\omega} = -\frac{1}{e}\sqrt{\frac{a(1-e^2)}{GM\mu^2}} \left[F_r \cos f + F_\theta \left(\xi\cos f - 2\sin f\right)\right], \tag{6.162}$$

$$\dot{f} = \sqrt{\frac{GM}{a^3}} \frac{(1 + e\cos f)^2}{(1-e^2)^{3/2}} - \dot{\omega}, \tag{6.163}$$

where

$$\xi = \frac{e \sin f}{1 + e\cos f}. \tag{6.164}$$

For a perturbation confined to the plane,

$$\boxed{i = \dot{\Omega} = 0.} \tag{6.165}$$

These equations are equations for the *osculating* elements. The quantities a, e, ω are therefore generally functions of time even though the actual position and momentum at each instant are represented by a Keplerian orbit.

6.15 Orbit-averaged evolution

The element rates above are instantaneous. For a weak perturbation, the long-term evolution is obtained by inserting the unperturbed Keplerian orbit into the right-hand sides and averaging over one orbital period.

For any quantity X,

$$\boxed{\left\langle \dot{X} \right\rangle = \frac{1}{P} \int_0^P \dot{X}\, dt.} \tag{6.166}$$

For an unperturbed Kepler ellipse,

$$L = \mu r^2 \dot{f}, \tag{6.167}$$

and therefore

$$\boxed{dt = \frac{\mu r^2}{L}\, df.} \tag{6.168}$$

To first order in the perturbing force, the distinction between $d\theta$ and df in this averaging relation produces only higher-order corrections.

The average rate can therefore be written as

$$\left\langle \dot{X} \right\rangle = \frac{1}{P} \int_0^{2\pi} \dot{X} \frac{\mu r^2}{L}\, df. \tag{6.169}$$

A nonzero value of $\left\langle \dot{X} \right\rangle$ represents a secular change. Terms whose integral over $0 \le f \le 2\pi$ vanishes describe only short-period oscillations.

6.16 Nearly circular orbits

As $e \to 0$, the argument of periapsis becomes undefined: a circle has no distinguished periapsis. The explicit $1/e$ in (6.162) is therefore a coordinate singularity.

Instead of (e, ω), use the two Cartesian components of the eccentricity vector:

$$\boxed{\epsilon_1 = e\cos\omega, \qquad \epsilon_2 = e\sin\omega.} \tag{6.170}$$

For small e, define

$$\Phi \equiv \theta\big|_{e=0}, \tag{6.171}$$

and write

$$\theta = \Phi + 2\epsilon_1 \sin\Phi - 2\epsilon_2\cos\Phi + O(e^2). \tag{6.172}$$

Using

$$r = \frac{a(1-e^2)}{1 + e\cos(\theta - \omega)} \tag{6.173}$$

and retaining terms through first order in ϵ_1, ϵ_2 gives

$$\boxed{r = a\left(1 - \epsilon_1\cos\Phi - \epsilon_2\sin\Phi\right) + O(e^2).} \tag{6.174}$$

Differentiating,

$$\dot{r} = an\left(\epsilon_1\sin\Phi - \epsilon_2\cos\Phi\right) + O(e^2), \tag{6.175}$$

where

$$n = \sqrt{\frac{GM}{a^3}}. \tag{6.176}$$

Also,

$$\boxed{\dot{\theta} = n\,(1 + 2\epsilon_1 \cos\Phi + 2\epsilon_2 \sin\Phi) + O(e^2).} \tag{6.177}$$

Define

$$\delta\theta \equiv \theta - \Phi = 2\epsilon_1 \sin\Phi - 2\epsilon_2 \cos\Phi. \tag{6.178}$$

The nonsingular variables ϵ_1, ϵ_2 are preferable to ω near circularity because they remain finite as $e \to 0$.

6.16.1 First-order equations for nearly circular motion

Substituting the small-e parameterization into the energy equation gives

$$\boxed{\dot{a} = \sqrt{\frac{4a^3}{GM\mu^2}}\,[\delta\theta\, F_r + (1 + \epsilon_1 \cos\Phi + \epsilon_2 \sin\Phi)\, F_\theta] + O(e^2 F).} \tag{6.179}$$

The eccentricity vector is

$$\vec{A} = GM\mu^2\left(\epsilon_1 \hat{\Omega} + \epsilon_2 \hat{y}\right) + O(e^2), \tag{6.180}$$

and its evolution gives

$$\boxed{\dot{\epsilon}_1 = \sqrt{\frac{a}{GM\mu^2}}\,[\sin\Phi\, F_r + 2\cos\Phi\, F_\theta + \delta\theta\,(2\cos\Phi\, F_r - 3\sin\Phi\, F_\theta)] + O(e^2 F),} \tag{6.181}$$

$$\boxed{\dot{\epsilon}_2 = \sqrt{\frac{a}{GM\mu^2}}\,[-\cos\Phi\, F_r + 2\sin\Phi\, F_\theta + \delta\theta\,(2\sin\Phi\, F_r + 3\cos\Phi\, F_\theta)] + O(e^2 F).} \tag{6.182}$$

To zeroth order in e,

$$\boxed{\dot{\Phi} = n + O(e^2).} \tag{6.183}$$

Using Φ as the independent variable,

$$\boxed{\frac{da}{d\Phi} = \frac{2a^3}{GM\mu}\,[\delta\theta\, F_r + (1 + \epsilon_1 \cos\Phi + \epsilon_2 \sin\Phi)\, F_\theta]\,,} \tag{6.184}$$

$$\boxed{\frac{d\epsilon_1}{d\Phi} = \frac{a^2}{GM\mu}\,[\sin\Phi\, F_r + 2\cos\Phi\, F_\theta + \delta\theta\,(2\cos\Phi\, F_r - 3\sin\Phi\, F_\theta)]\,,} \tag{6.185}$$

$$\boxed{\frac{d\epsilon_2}{d\Phi} = \frac{a^2}{GM\mu}\,[-\cos\Phi\, F_r + 2\sin\Phi\, F_\theta + \delta\theta\,(2\sin\Phi\, F_r + 3\cos\Phi\, F_\theta)]\,.} \tag{6.186}$$

6.17 Worked example: radial inverse-cube perturbation

Consider the perturbing force

$$\boxed{\vec{F} = \frac{\alpha}{r^3}\hat{r}.} \tag{6.187}$$

Thus

$$F_r = \frac{\alpha}{r^3}, \qquad F_\theta = 0. \tag{6.188}$$

Rate of the semimajor axis

Differentiating (**??**), we obtain

$$\dot{E} = \frac{GM\mu}{2a^2}\dot{a}. \tag{6.189}$$

The exact rate of change of the energy of the actual orbit is

$$\dot{E} = \vec{F} \cdot \vec{v}. \tag{6.190}$$

Since the perturbation is radial,

$$\vec{F} = \frac{\alpha}{r^3}\hat{r}, \tag{6.191}$$

and

$$\vec{v} = \dot{r}\,\hat{r} + r\dot{f}\,\hat{\phi}, \tag{6.192}$$

we have

$$\dot{E} = \frac{\alpha}{r^3}\dot{r}. \tag{6.193}$$

For the osculating Keplerian orbit,

$$\dot{r} = \frac{GM\mu}{L}e\sin f. \tag{6.194}$$

Therefore,

$$\dot{E} = \frac{\alpha GM\mu}{Lr^3}e\sin f. \tag{6.195}$$

Comparing this result with (6.191),

$$\frac{GM\mu}{2a^2}\dot{a} = \frac{\alpha GM\mu}{Lr^3}e\sin f, \tag{6.196}$$

so that

$$\boxed{\dot{a} = \frac{2\alpha a^2 e\sin f}{Lr^3}}. \tag{6.197}$$

Rate of the eccentricity

Define the eccentricity vector by

$$\vec{A} = \vec{p} \times \vec{L} - GM\mu^2\hat{r}. \tag{6.198}$$

For the osculating Keplerian orbit,

$$A = GM\mu^2 e. \tag{6.199}$$

Since the perturbing force is radial,

$$\dot{\vec{L}} = \vec{r} \times \vec{F} = \vec{r} \times \frac{\alpha}{r^3}\hat{r} = 0. \tag{6.200}$$

Hence

$$\dot{L} = 0. \tag{6.201}$$

Differentiating

$$L^2 = GM\mu^2 a(1 - e^2), \tag{6.202}$$

gives

$$2L\dot{L} = GM\mu^2\left[(1-e^2)\dot{a} - 2ae\dot{e}\right]. \tag{6.203}$$

Since $\dot{L} = 0$,

$$(1-e^2)\dot{a} = 2ae\dot{e}. \tag{6.204}$$

Therefore,

$$\dot{e} = \frac{1-e^2}{2ae}\dot{a}. \tag{6.205}$$

Substituting (6.199),

$$\dot{e} = \frac{1-e^2}{2ae}\frac{2\alpha a^2 e\sin f}{Lr^3}. \tag{6.206}$$

Thus,

$$\boxed{\dot{e} = \frac{\alpha a(1-e^2)\sin f}{Lr^3}}. \tag{6.207}$$

Rate of the argument of periapsis

We now determine the rate of the direction of the eccentricity vector.

Differentiating (6.200),

$$\dot{\vec{A}} = \dot{\vec{p}}\times\vec{L} + \vec{p}\times\dot{\vec{L}} - GM\mu^2\dot{\hat{r}}. \tag{6.208}$$

Since $\dot{\vec{L}} = 0$,

$$\dot{\vec{A}} = \dot{\vec{p}}\times\vec{L} - GM\mu^2\dot{\hat{r}}. \tag{6.209}$$

The momentum equation is

$$\dot{\vec{p}} = -\frac{GM\mu}{r^2}\hat{r} + \frac{\alpha}{r^3}\hat{r}. \tag{6.210}$$

Therefore,

$$\dot{\vec{A}} = \left(-\frac{GM\mu}{r^2} + \frac{\alpha}{r^3}\right)\hat{r}\times\vec{L} - GM\mu^2\dot{\hat{r}}. \tag{6.211}$$

Now,

$$\vec{L} = L\hat{L}, \tag{6.212}$$

and

$$\hat{r}\times\hat{L} = -\hat{\phi}, \tag{6.213}$$

so that

$$\hat{r}\times\vec{L} = -L\hat{\phi}. \tag{6.214}$$

Also,

$$\dot{\hat{r}} = \dot{f}\,\hat{\phi}. \tag{6.215}$$

Since

$$\dot{f} = \frac{L}{\mu r^2}, \tag{6.216}$$

we have

$$GM\mu^2\dot{\hat{r}} = \frac{GM\mu L}{r^2}\hat{\phi}. \tag{6.217}$$

Consequently, the terms containing the central gravitational force cancel, leaving

$$\dot{\vec{A}} = -\frac{\alpha L}{r^3}\hat{\phi}. \tag{6.218}$$

Let $\hat{\boldsymbol{e}}$ be the unit vector pointing toward periapsis and let $\hat{\boldsymbol{e}}_\perp$ be the unit vector obtained by rotating $\hat{\boldsymbol{e}}$ by 90° in the direction of increasing true anomaly. Then

$$\vec{A} = A\hat{\boldsymbol{e}}, \tag{6.219}$$

and therefore

$$\dot{\vec{A}} = \dot{A}\,\hat{\boldsymbol{e}} + A\dot{\omega}\,\hat{\boldsymbol{e}}_\perp. \tag{6.220}$$

The radial and transverse unit vectors are related to the periapsis basis by

$$\hat{r} = \cos f\,\hat{\boldsymbol{e}} + \sin f\,\hat{\boldsymbol{e}}_\perp, \tag{6.221}$$

and

$$\hat{\phi} = -\sin f\,\hat{\boldsymbol{e}} + \cos f\,\hat{\boldsymbol{e}}_\perp. \tag{6.222}$$

Substituting the latter relation into (6.220),

$$\dot{\vec{A}} = -\frac{\alpha L}{r^3}\left(-\sin f\,\hat{\boldsymbol{e}} + \cos f\,\hat{\boldsymbol{e}}_\perp\right). \tag{6.223}$$

Hence,

$$\dot{\vec{A}} = \frac{\alpha L}{r^3}\sin f\,\hat{\boldsymbol{e}} - \frac{\alpha L}{r^3}\cos f\,\hat{\boldsymbol{e}}_\perp. \tag{6.224}$$

Comparing this with (6.222), we obtain

$$\dot{A} = \frac{\alpha L}{r^3}\sin f, \tag{6.225}$$

and

$$A\dot{\omega} = -\frac{\alpha L}{r^3}\cos f. \tag{6.226}$$

Using

$$A = GM\mu^2 e, \tag{6.227}$$

we finally obtain

$$\boxed{\dot{\omega} = -\frac{\alpha L}{GM\mu^2 e\,r^3}\cos f}. \tag{6.228}$$

6.17.1 Semi-major axis

Equation (6.160) gives

$$\dot{a} = \sqrt{\frac{4a^3}{GM\mu^2(1-e^2)}}\,e\sin f\,\frac{\alpha}{r^3}. \tag{6.229}$$

Hence

$$\boxed{\dot{a} = \frac{2\alpha a^2}{GM\mu r^3}\dot{r}.} \tag{6.230}$$

For the Keplerian ellipse introduce the eccentric anomaly u:

$$r = a(1 - e\cos u), \tag{6.231}$$

and

$$\dot r = \frac{ane\sin u}{1 - e\cos u}, \qquad n = \sqrt{\frac{GM}{a^3}}. \tag{6.232}$$

Therefore

$$\boxed{\dot a = \frac{2\alpha en\sin u}{GM\mu(1 - e\cos u)^4}.} \tag{6.233}$$

Over one unperturbed orbit,

$$dt = \frac{1 - e\cos u}{n}\,du, \tag{6.234}$$

so

$$\begin{aligned}\Delta a &= \int_0^{2\pi} \dot a\,dt && (6.235)\\ &= \frac{2\alpha e}{GM\mu}\int_0^{2\pi} \frac{\sin u}{(1 - e\cos u)^3}\,du && (6.236)\\ &= 0. && (6.237)\end{aligned}$$

Thus

$$\boxed{\langle \dot a\rangle = 0} \tag{6.238}$$

to first order in the perturbation.

The osculating semi-major axis therefore oscillates during an orbit but has no secular drift.

6.17.2 Eccentricity

From angular-momentum conservation,

$$L^2 = GM\mu^2 a(1 - e^2) \tag{6.239}$$

and therefore

$$0 = (1 - e^2)\dot a - 2ae\dot e. \tag{6.240}$$

Hence

$$\boxed{\dot e = \frac{1 - e^2}{2ae}\dot a.} \tag{6.241}$$

Since $\langle \dot a\rangle = 0$,

$$\boxed{\langle \dot e\rangle = 0.} \tag{6.242}$$

6.17.3 Apsidal precession

For the radial force, (6.162) reduces to

$$\boxed{\dot\omega = -\frac{\alpha L\cos f}{GM\mu^2 e\, r^3}.} \tag{6.243}$$

To first order in α, evaluate the secular change using the unperturbed Keplerian relation

$$r = \frac{\lambda}{1 + e\cos f}, \qquad \lambda = a(1 - e^2), \tag{6.244}$$

together with

$$dt = \frac{\mu r^2}{L}\, df. \tag{6.245}$$

Then

$$\begin{aligned}
\Delta\omega &= \int_0^{2\pi} \dot{\omega}\, dt && (6.246)\\
&= -\frac{\alpha}{GM\mu e}\int_0^{2\pi} \frac{\cos f}{r}\, df && (6.247)\\
&= -\frac{\alpha}{GM\mu e}\int_0^{2\pi} \frac{1+e\cos f}{\lambda}\cos f\, df && (6.248)\\
&= -\frac{\alpha}{GM\mu e\lambda}\left[\int_0^{2\pi}\cos f\, df + e\int_0^{2\pi}\cos^2 f\, df\right]. && (6.249)
\end{aligned}$$

Since

$$\int_0^{2\pi}\cos f\, df = 0, \qquad \int_0^{2\pi}\cos^2 f\, df = \pi, \tag{6.250}$$

we obtain

$$\boxed{\Delta\omega = -\frac{\pi\alpha}{GM\mu\, a(1-e^2)}.} \tag{6.251}$$

Using

$$P = 2\pi\sqrt{\frac{a^3}{GM}}, \tag{6.252}$$

the secular rate is

$$\boxed{\langle\dot{\omega}\rangle = -\frac{\alpha}{2\mu\sqrt{GM}\, a^{5/2}(1-e^2)}.} \tag{6.253}$$

The same result follows directly from the exact Binet equation for the full force

$$F_r = -\frac{GM\mu}{r^2} + \frac{\alpha}{r^3}. \tag{6.254}$$

Writing $u = 1/r$ gives

$$u'' + \left(1 + \frac{\alpha\mu}{L^2}\right)u = \frac{GM\mu^2}{L^2}, \tag{6.255}$$

where primes denote derivatives with respect to θ. Hence the radial oscillation has angular frequency

$$\kappa = \sqrt{1 + \frac{\alpha\mu}{L^2}}, \tag{6.256}$$

and the apsidal angle is

$$\Delta\theta_{\text{apsis}} = \frac{2\pi}{\kappa}. \tag{6.257}$$

The exact apsidal shift relative to 2π is therefore

$$\Delta\omega_{\text{exact}} = 2\pi\left[\frac{1}{\sqrt{1+\alpha\mu/L^2}} - 1\right]. \tag{6.258}$$

For a weak perturbation,

$$\frac{\alpha\mu}{L^2} \ll 1, \tag{6.259}$$

so

$$\Delta\omega_{\rm exact} = -\frac{\pi\alpha\mu}{L^2} + O(\alpha^2) \tag{6.260}$$
$$= -\frac{\pi\alpha}{GM\mu\, a(1-e^2)} + O(\alpha^2), \tag{6.261}$$

which agrees with the orbit-averaged planetary equation.

6.18 Rømer delay for a Keplerian binary

For a binary of component masses m_1 and m_2,

$$M = m_1 + m_2. \tag{6.262}$$

Let $\vec{r}$ be the relative separation and let the primary's barycentric position be

$$\vec{x}_1 = \frac{m_2}{M}\vec{r}. \tag{6.263}$$

If $\hat{R}$ points from the source towards the observer, the light-travel-time contribution is

$$\Delta_R = -\frac{\vec{x}_1 \cdot \hat{R}}{c}. \tag{6.264}$$

Using the orbital-plane basis,

$$\vec{r} = r\left[\cos\theta\,\hat{\Omega} + \sin\theta\,\hat{y}\right], \tag{6.265}$$

with

$$\theta = \omega + f, \tag{6.266}$$

and

$$\hat{y}\cdot\hat{R} = \sin\iota, \qquad \hat{\Omega}\cdot\hat{R} = 0, \tag{6.267}$$

we obtain

$$\Delta_R = \frac{m_2}{Mc}\, r\sin\iota\,\sin(\omega+f). \tag{6.268}$$

Define

$$a_1 = \frac{m_2}{M}a, \tag{6.269}$$

and use the eccentric-anomaly relations

$$r\cos f = a(\cos u - e), \tag{6.270}$$
$$r\sin f = a\sqrt{1-e^2}\,\sin u. \tag{6.271}$$

Since

$$\sin(\omega+f) = \sin\omega\cos f + \cos\omega\sin f, \tag{6.272}$$

we obtain

$$\boxed{\Delta_R = \frac{a_1\sin\iota}{c}\left[(\cos u - e)\sin\omega + \sqrt{1-e^2}\,\sin u\cos\omega\right].} \tag{6.273}$$

Chapter 7

Interstellar Medium